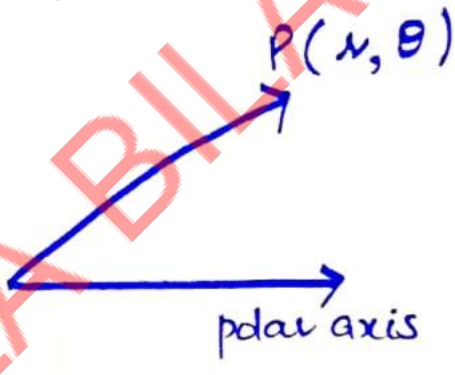
Lecture 23

Polar Coordinate System

Take a point O called origin/pole & draw a ray called polar axis.

Now take a point P with coordinates (r, θ) where "r" is distance of $\overline{OP}$ & θ is angle b/w $\overline{OP}$ & polar axis.



This is how we form a polar coordinate System.

Radial distance:-

The number r is called radial distance of P.

Polar angle:-

θ is called the polar angle of P.

P is written as $P(r, \theta)$ (e.g) $(6, 45^\circ), (4, 330^\circ)$ are plotted in polar coordinate System.

Remark. The points $(3, 225^\circ)$ & $(-3, 45^\circ)$ are same. (-ve sign shows that point is one extension of angle's terminal side rather than on terminal side itself. As $(\theta + 180)$ is extension of θ. So $(-r, \theta)$ can also be written as $(r, \theta + 180)$

NOTES BY KINZA BILAL

Scanned with CamScanner

Relation b/w Polar & Rectangular coordinates:-

Polar coordinates $P(x, y)$

Rectangular Coordinates $P(r, \theta)$

$x = r\cos\theta$ & $y = r\sin\theta$

Conversion:-

From polar to Cartesian/Rectangular:-

$x = r\cos\theta$ & $y = r\sin\theta$

From Rectangular to Polar:-

$r^2 = x^2 + y^2$ & $\tan\theta = \frac{y}{x}$

Example:- Find rectangular coordinates of P whose polar coordinates are $(6, 135^\circ)$

$r = 6$, $\theta = 135^\circ$

As $x = r\cos\theta$

$= 6(\cos 135)$

$= 6\left(-\frac{\sqrt{2}}{2}\right)$

$= -3\sqrt{2}$

$y = r\sin\theta$

$y = 6\sin 135^\circ$

$= 6\left(\frac{\sqrt{2}}{2}\right)$

$y = 3\sqrt{2}$

So rectangular coordinates of P are

$(-3\sqrt{2}, 3\sqrt{2})$

NOTES BY KINZA BILAL

Scanned with CamScanner

Example:- Find polar coordinates of P whose rectangular coordinates are $(-2, 2\sqrt{3})$.

here $x = -2$ & $y = 2\sqrt{3}$

$$r^2 = x^2 + y^2$$
$$= (-2)^2 + (2\sqrt{3})^2$$
$$= 4 + 4(3)$$
$$= 4 + 12$$
$$r^2 = 16$$
$$r = 4$$

$$\tan\theta = \frac{y}{x}$$
$$\tan\theta = \frac{2\sqrt{3}}{-2}$$
$$\tan\theta = (-\sqrt{3})$$
$$\theta = \tan^{-1}(-\sqrt{3})$$
$$\theta = \frac{2\pi}{3} \text{ or } 120^\circ$$

So the polar coordinates are (r, θ) or $(-r, \theta + 180)$

(i-e) $\left(4, \frac{2\pi}{3}\right), \left(-4, \frac{5\pi}{3}\right)$

Lines in Polar coordinates:-

If we have $x = a$ (i-e) point $(a, 0)$, the equation $x = r\cos\theta$ will be $a = r\cos\theta$.

If we have $y = b$ (i.e) point $(0, b)$, the equation $y = r\sin\theta$ will be $b = r\sin\theta$.

If $\theta = \theta_0$ it means line passes through origin.

As general equation of line is $Ax + By + C = 0$.

Put values:- $A(r\cos\theta) + B(r\sin\theta) + C = 0$

$$r(A\cos\theta + B\sin\theta) + C = 0$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Circles in polar coordinates:-

If we have a circle of radius 'a' & its center is (r_0, θ_0), then by applying law of cosine

(general form:- $a^2 = b^2 + c^2 - 2bc\cos\theta$)

$$a^2 = r^2 - 2rr_0\cos(\theta - \theta_0) + r_0^2 \quad \text{---} \quad (1)$$

Special Cases:-

⇒ A circle of radius a & center at origin.

Let $r_0 = 0$ (put in eq 1)

$$a^2 = r^2 - 2r(0)\cos(\theta - \theta_0) + 0^2$$

$$a^2 = r^2$$

This circle consists of all points $P(r, \theta)$ for which $r = a$, regardless of θ.

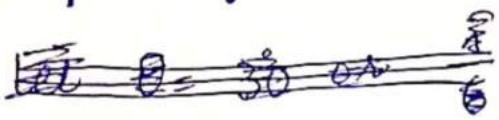
⇒ If circle of radius a & center on x-axis & passes Through origin, Then polar coordinates of center are either $(a, 0)$ or (a, π) depending on whether center is on right or left of origin.

NOTES BY KINZA BILAL

Scanned with CamScanner

Lecture 24 Sketching

graph of the curve having equ $r = \sin\theta$



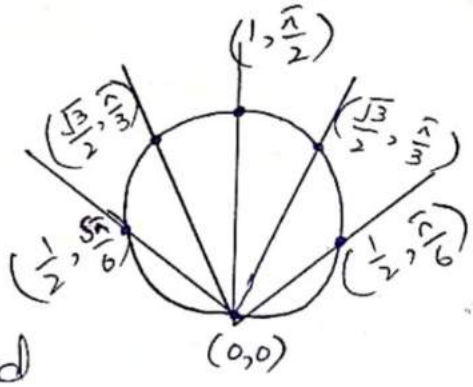
increase θ by 30° or $\frac{\pi}{6}$.

θ (radians)	0	$\frac{\pi}{6}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{5\pi}{6}$
$r = \sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{1}{2}$

θ	π	$\frac{7\pi}{6}$	$\frac{4\pi}{3}$	$\frac{3\pi}{2}$	$\frac{5\pi}{3}$	$\frac{11\pi}{6}$	2π
$r = \sin\theta$	0	$-\frac{1}{2}$	$-\frac{\sqrt{3}}{2}$	-1	$-\frac{\sqrt{3}}{2}$	$-\frac{1}{2}$	0

There are 13 pairs in Table but only 6 points are plotted bcz others are same as them.

(-ve value of r & θ is increased by 180° or π)



Equation of This circle:- As $r = \sin\theta$

(multiply b/s by r) $\quad r^2 = r\sin\theta$

($r^2 = x^2 + y^2$ & $y = r\sin\theta$) $\quad x^2 + y^2 = y$

$x^2 + y^2 - y = 0$

completing square $\quad x^2 + \left[y^2 - 2y\left(\frac{1}{2}\right) + \left(\frac{1}{2}\right)^2\right] - \left(\frac{1}{2}\right)^2 = 0$

$x^2 + \left(y - \frac{1}{2}\right)^2 = \frac{1}{4}$

NOTES BY KINZA BILAL

Scanned with CamScanner

Scanned with CamScanner

$$x^2 + \left(y - \frac{1}{2}\right)^2 = \frac{1}{4} \quad \text{——— (1)}$$

As general eqn of circle is

$$(x - x_0)^2 + (y - y_0)^2 = r^2 \quad \left[(x_0, y_0) \text{ is center } \& \ r \text{ is radius}\right]$$

In eq (1) $x_0 = 0$ & $y_0 = -\frac{1}{2}$ | $r^2 = \frac{1}{4}$, $r = \frac{1}{2}$

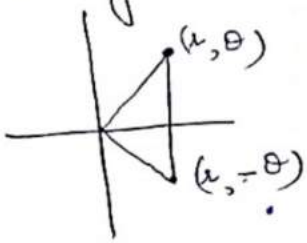
So center (x_0, y_0) is $\boxed{\left(0, -\frac{1}{2}\right)}$ & $\boxed{r = \frac{1}{2}}$

Curves in Polar Coordinates:-

1. Symmetry

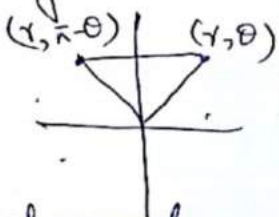
(i) about x-axis/ initial line:-

If equation of a line remains unchanged when (r, θ) is replaced by $(r, -\theta)$.



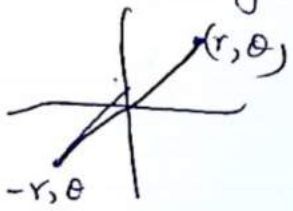
(ii) about y-axis:-

If equation of a line remains unchanged when (r, θ) is replaced by $(r, \pi - \theta)$



(iii) about Pole:-

If the equation of a line remains unchanged when (r, θ) is replaced by $(-r, \theta)$



NOTES BY KINZA BILAL

Scanned with CamScanner

Scanned with CamScanner

2. Position of the pole Relative to curve:-

Put $r=0$ & find value of θ for position.

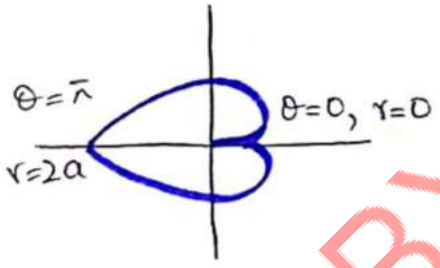
3. Table of values:-

Construct a sufficiently complete table of values.

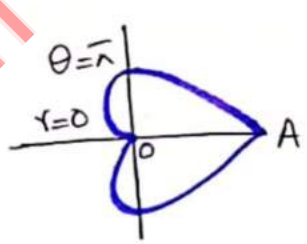
Position of Pole relative to curve:-

When $r=0$ & $\theta=0$ Then the curve passes through the pole.

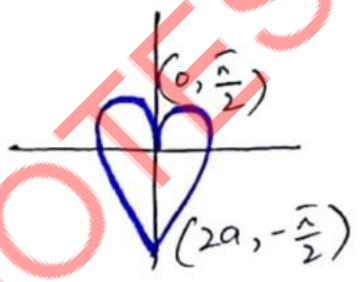
Cardioids:-



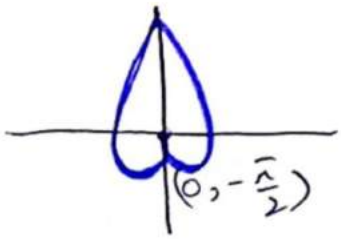
$r = a(1-\cos\theta)$



$r = a(1+\cos\theta)$



$r = a(1-\sin\theta)$



$r = a(1+\sin\theta)$

NOTES BY KINZA BILAL

Scanned with CamScanner

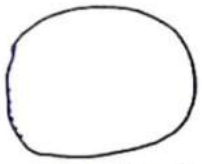
Scanned with CamScanner

Cardidoids & Limacons:-

$r = a + b\sin\theta$, $r = a - b\sin\theta$

$r = a + b\cos\theta$, $r = a - b\cos\theta$

These equation form polar curves called Limacons. If $a = b$ then above equations become same as for cardidoids. Their position depends upon whether $\sin\theta$ or $\cos\theta$ is +ve or -ve.



$\frac{a}{b} \geq 2$



$\frac{a}{b} < 1$

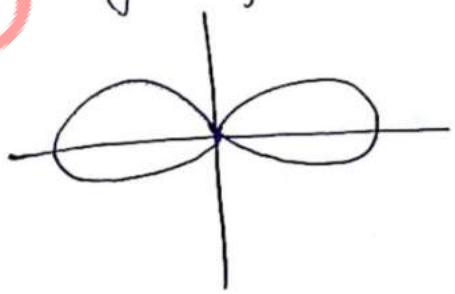
Lemicon with inner loop.

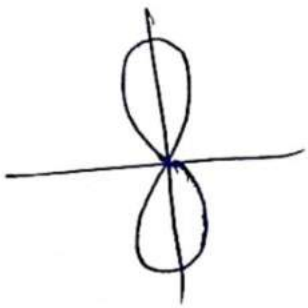
Leminscate:- if $a \neq 0$, then equation of form

$r^2 = a^2\cos 2\theta$, $r^2 = -a^2\cos 2\theta$

$r^2 = a^2\sin 2\theta$, $r^2 = -a^2\sin 2\theta$

represent propeller-shaped curves, called Leminscate. These are centered at origin. But position depends on sign of a^2.





(e.g) $r^2 = 4\cos 2\theta$ is eqn. of a leminscate.

NOTES BY KINZA BILAL

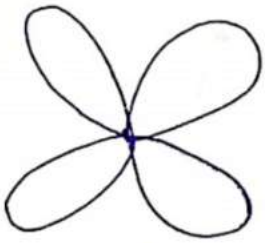
Scanned with CamScanner

Scanned with CamScanner

Rose Curves:-

Equations of the form ~~$r = a \sin\theta$~~ $r = a \sin n\theta$ & $r = a \cos n\theta$ represent flower shaped curves called Roses.

The rose has n equally spaced petals or loops if n is odd & $2n$ equally spaced petals if n is even.



($n=2$, four petals)



($n=3$, Three petals)

orientation depends upon sign of a. & if $\sin\theta$ or $\cos\theta$ appears in equation.

Spiral:-

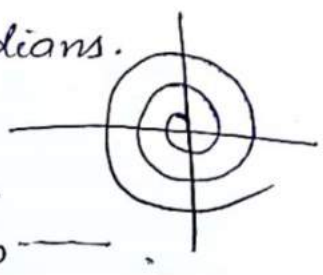
A curve that winds around the origin infinitely many times in such a way that r increases/decreases steadily as θ.

It has equations of form

$r = a\theta \quad (\theta \geq 0)$ or $r = a\theta \quad (\theta \leq 0)$

here a is +ve & θ is in radians.

its intersection with x-axis occurs at $\theta = 0, \pi, 2\pi, 3\pi, \ldots$ & intersection with y-axis occurs at $\theta = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2}, \frac{7\pi}{2}, \ldots$



NOTES BY KINZA BILAL

Scanned with CamScanner

Scanned with CamScanner

Lecture 25

Double integral in Polar Coordinates

Double integral in Polar coordinates is given as

$$\iint_R f(r,\theta)\,dA = \int_{\theta=\alpha}^{\theta=\beta}\int_{r=r_1(\theta)}^{r=r_2(\theta)} f(r,\theta)\,r\,dr\,d\theta$$

Example:- Find limits of integration for integrating $f(r,\theta)$ over Region R that lies inside the cardioid $r=1+\cos\theta$ & outside the circle $r=1$.

As

$$\iint_R f(r,\theta)\,dA = \int_{\theta=\alpha}^{\theta=\beta}\int_{r=r_1(\theta)}^{r=r_2(\theta)} f(r,\theta)\,r\,dr\,d\theta$$

$r = 1+\cos\theta$

put $r=1$

$1 = 1+\cos\theta$

$1-1 = \cos\theta$

$0 = \cos\theta$

$\cos^{-1}(0) = \theta$

$\pm\frac{\pi}{2} = \theta$

So limits are

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\int_{1}^{1+\cos\theta} f(r,\theta)\,r\,dr\,d\theta.$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Scanned with CamScanner

Example:- Evaluate $\iint_R \sin\theta \, dA$

where R is region in first quad. That is outside the circle $r=2$ & inside cardioid $r = 2(1+\cos\theta)$

As $r = 2(1+\cos\theta)$

put $r = 2$

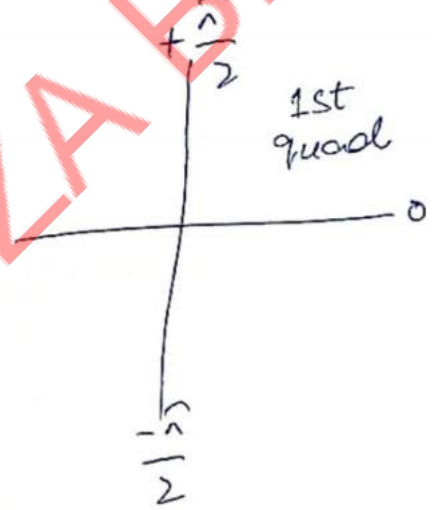
$2 = 2(1+\cos\theta)$

$1 = 1+\cos\theta$

$0 = \cos\theta$

$\cos^{-1}(0) = \theta$

$\pm \frac{\pi}{2} = \theta$



So $\int_0^{\pi/2} \int_2^{2(1+\cos\theta)} \sin\theta \, r \, dr \, d\theta$

$= \int_0^{\frac{\pi}{2}} \left[\sin\theta \, \frac{r^2}{2} \right]_2^{2(1+\cos\theta)} d\theta$

$= \int_0^{\frac{\pi}{2}} \left[\left(\sin\theta \, \frac{(2(1+\cos\theta))^2}{2} \right) - \left(\sin\theta \left(\frac{(2)^2}{2} \right) \right) \right] d\theta$

$= \int_0^{\frac{\pi}{2}} \left[\left(\sin\theta \left(\frac{4(1+\cos\theta)^2}{2} \right) \right) - \left(\sin\theta \, \frac{4}{2} \right) \right] d\theta$

NOTES BY KINZA BILAL

Scanned with CamScanner

Scanned with CamScanner

$$= \int_0^{\frac{\pi}{2}} \left[2\sin\theta(1+\cos\theta)^2 - 2\sin\theta \right] d\theta$$

$$= 2\int_0^{\frac{\pi}{2}} \left(-\sin\theta(1+\cos\theta)^2 + \sin\theta \right) d\theta$$

$$= 2\left[-\frac{(1+\cos\theta)^3}{3} + \cos\theta \right]_0^{\frac{\pi}{2}} + C$$

$$= 2\left[\left(-\frac{(1+\cos\frac{\pi}{2})^3}{3} + \cos\frac{\pi}{2} \right) - \left(-\frac{(1+\cos 0^\circ)^3}{3} + \cos 0^\circ \right) \right]$$

$$= 2\left[\left(-\frac{1}{3} + 0 \right) - \left(-\frac{8}{3} + 1 \right) \right]$$

$$= 2\left(-\frac{1}{3} + \frac{8}{3} - \frac{1}{1} \right)$$

$$= 2\left(\frac{-1+8-3}{3} \right) \Rightarrow 2\left(\frac{4}{3} \right)$$

$$= \frac{8}{3} \quad \text{Ans.}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Scanned with CamScanner

Example:- Use a double polar integral to find area enclosed by three-petaled rose $r = \sin 3\theta$. We calculate are in 1st quad & multiply by 3.

$$A = 3\iint_R dA$$

$$A = 3\iint_R r\,dr\,d\theta$$

$$= 3\int_0^{\frac{\pi}{3}}\int_0^{\sin 3\theta} r\,dr\,d\theta$$

$$= 3\int_0^{\frac{\pi}{3}}\left[\frac{r^2}{2}\right]_0^{\sin 3\theta} d\theta \Rightarrow \frac{3}{2}\int_0^{\frac{\pi}{3}} r^2\Big|_0^{\sin 3\theta} d\theta$$

$$= \frac{3}{2}\int_0^{\pi/3}\left((\sin^2 3\theta) - 0^2\right) d\theta$$

$$= \frac{3}{2}\int_0^{\frac{\pi}{3}} \frac{1-\cos 6\theta}{2}\, d\theta$$

$$= \frac{3}{2}\int_0^{\frac{\pi}{3}}\left(\frac{1}{2} - \frac{\cos 6\theta}{2}\right) d\theta \Rightarrow \frac{3}{4}\int_0^{\frac{\pi}{3}}(1-\cos 6\theta)\,d\theta$$

$$A = \frac{3}{4}\left(\theta - \frac{\sin 6\theta}{6}\right)\Big|_0^{\frac{\pi}{3}} \Rightarrow \frac{3}{4}\left(\frac{\pi}{3} - \frac{\sin 6\left(\frac{\pi}{3}\right)}{6}\right)$$

$$A = \frac{3}{4}\left(\frac{\pi}{3} - \frac{\sin 2\pi}{6}\right) \Rightarrow \frac{3}{4}\left(\frac{\pi}{3}\right) \Rightarrow \boxed{\frac{\pi}{4}}$$ Ans.

$r = \sin 3\theta$

put $r = 0$

$0 = \sin 3\theta$

$\sin^{-1}(0) = 3\theta$

$\pi = 3\theta$

$\frac{\pi}{3} = \theta$

$\because \sin^2\theta = \frac{1-\cos 2\theta}{2}$

NOTES BY KINZA BILAL

Scanned with CamScanner

Scanned with CamScanner

Find area enclosed by leminscate $r^2 = 4\cos 2\theta$. Total area is four times the first quad portion.

$r = 0$, $r^2 = 4\cos 2\theta$

$r = \sqrt{4\cos 2\theta}$

$0 = 4\cos 2\theta$

$0 = \cos 2\theta$

$\cos^{-1}(0) = 2\theta$

$\frac{\pi}{2} = \theta$

$\frac{\pi}{4} = \theta$, $\theta = 0$

$$A = 4\int_0^{\pi/4}\int_0^{\sqrt{4\cos 2\theta}} r\,dr\,d\theta \Rightarrow 4\int_0^{\frac{\pi}{4}} \frac{r^2}{2}\Big|_0^{\sqrt{4\cos 2\theta}} d\theta$$

$$= \frac{4}{2}\int_0^{\pi/4} \left(\sqrt{4\cos 2\theta}\right)^2 - 0^2 \; d\theta \Rightarrow 2\int_0^{\frac{\pi}{4}} 4\cos 2\theta\, d\theta$$

$$= 8\int_0^{\pi/4} \cos 2\theta\, d\theta \Rightarrow 8\left|\frac{\sin 2\theta}{2}\right|_0^{\frac{\pi}{4}}$$

$$= \frac{8}{2}\left(\sin 2\left(\frac{\pi}{4}\right) - \sin 2(0)\right)$$

$$= 4\left(\sin\frac{\pi}{2} - \sin 0\right)$$

$$= 4(1-0)$$

$$= 4(1)$$

$$\boxed{A = 4}$$ Ans.

NOTES BY KINZA BILAL

Scanned with CamScanner

Scanned with CamScanner

Changing Cartesian integral to polar.

Step 1: put $x = r\cos\theta$ & $y = r\sin\theta$ & replace $dx\,dy$ by $dr\,d\theta$.

Step 2: Supply polar limits of integration for R.

The integral becomes

$$\iint_R f(x,y)\,dx\,dy = \iint_{R'} f(r\cos\theta, r\sin\theta)\, r\,dr\,d\theta$$

Example:- Evaluate $\int_0^1 \int_0^{\sqrt{1-x^2}} (x^2+y^2)\,dy\,dx$ by changing into polar coordinates.

$0 \le y \le \sqrt{1-x^2}$ & $0 \le x \le 1$

$y = \sqrt{1-x^2}$ $\qquad \theta = 0$ & $\theta = \frac{\pi}{2}$

$y^2 = (\sqrt{1-x^2})^2$

$y^2 = 1 - x^2$

$x^2 + y^2 = 1$

here $r^2 = 1$

so $r = 1$

NOTES BY KINZA BILAL

Scanned with CamScanner

Scanned with CamScanner

$$\int_0^{\frac{\pi}{2}}\int_0^1 (r^2\cos^2\theta + r^2\sin^2\theta)\, r\, dr\, d\theta$$

$$= \int_0^{\frac{\pi}{2}}\int_0^1 r^2(\cos^2\theta + \sin^2\theta)\, r\, dr\, d\theta$$

$$= \int_0^{\frac{\pi}{2}}\int_0^1 r^2(1)\, r\, dr\, d\theta$$

$$= \int_0^{\pi/2}\int_0^1 r^3\, dr\, d\theta \Rightarrow \int_0^{\frac{\pi}{2}} \frac{r^4}{4}\Big|_0^1 d\theta$$

$$= \frac{1}{4}\int_0^{\pi/2}(1^4 - 0^4)\, d\theta \Rightarrow \frac{1}{4}\int_0^{\pi/2}(1)\, d\theta$$

$$= \frac{1}{4}|\theta|_0^{\frac{\pi}{2}} \Rightarrow \frac{1}{4}\left(\frac{\pi}{2} - 0\right)$$

$$= \frac{1}{4}\left(\frac{\pi}{2}\right) \Rightarrow \frac{\pi}{8}$$ Ans.

Example:- Evaluate $\iint_D \frac{dx\,dy}{x^2+y^2}$ by changing to polar coordinates where D is region of first quad b/w circles $x^2+y^2 = a^2$ & $x^2+y^2 = b^2$ $(0<a<b)$

NOTES BY KINZA BILAL

Scanned with CamScanner

Scanned with CamScanner

As $x^2+y^2=r^2$ if $a^2 \le r^2 \le b^2$

$a \le r \le b$

As quad is 1st So

$\theta = 0$ & $\theta = \frac{\pi}{2}$

$$= \int_0^{\frac{\pi}{2}} \int_a^b \frac{r\,dr\,d\theta}{r^2\cos^2\theta + r^2\sin^2\theta}$$

$$= \int_0^{\frac{\pi}{2}} \int_a^b \frac{r}{r^2}\,dr\,d\theta \Rightarrow \int_0^{\frac{\pi}{2}} \int_a^b \frac{1}{r}\,dr\,d\theta$$

$$= \int_0^{\frac{\pi}{2}} |\ln r|_a^b\,d\theta \Rightarrow \int_0^{\pi/2} (\ln b - \ln a)\,d\theta$$

$$= \int_0^{\frac{\pi}{2}} \ln\left(\frac{b}{a}\right) d\theta \Rightarrow \ln\left(\frac{b}{a}\right) \int_0^{\pi/2} d\theta$$

$$= \ln\left(\frac{b}{a}\right) |\theta|_0^{\frac{\pi}{2}} \Rightarrow \ln\left(\frac{b}{a}\right)\left(\frac{\pi}{2} - 0\right)$$

$$= \frac{\pi}{2} \ln\left(\frac{b}{a}\right) \quad \text{Ans.}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

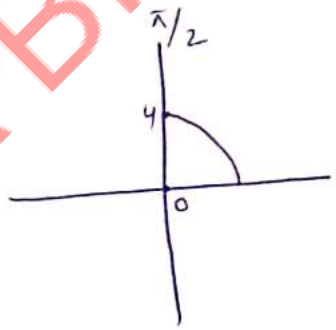
Scanned with CamScanner

Lecture No. 26

Examples:-

Example:- Evaluate $I = \int_0^4 \int_0^{\sqrt{4y-y^2}} (x^2+y^2)\, dx\, dy$ by changing into polar coordinates.

$0 \le x \le \sqrt{4y-y^2}$ & $0 \le y \le 4$



Take $x = \sqrt{4y-y^2}$

Take square on b/s

$x^2 = 4y - y^2$

$x^2 + y^2 = 4y$ (as general eqn. of circle is $x^2+y^2 = r^2$)

So here

$r^2 = 4y$

$r^2 = 4r\sin\theta$ $(\because y = r\sin\theta)$

$r = 4\sin\theta$ & $r = 0$

$\theta = 0$ & $\theta = \frac{\pi}{2}$

Now $I = \int_0^{\frac{\pi}{2}} \int_0^{4\sin\theta} r^2 \, r\, dr\, d\theta \Rightarrow \int_0^{\pi/2} \int_0^{4\sin\theta} r^3\, dr\, d\theta$

$= \int_0^{\pi/2} \left| \frac{r^4}{4} \right|_0^{4\sin\theta} d\theta \Rightarrow \int_0^{\pi/2} \left(\frac{(4\sin\theta)^4}{4} - \frac{0^4}{4} \right) d\theta$

$= \int_0^{\pi/2} \frac{4^4 \sin^4\theta}{4}\, d\theta$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$= \int_0^{\pi/2} 4^3 \sin^4\theta \, d\theta$$

$$= \int_0^{\pi/2} 64 \sin^4\theta \, d\theta$$

Using Walli's formula

$$= \int_0^{\pi/2} 64 \frac{(4-1)(4-3)}{4(4-2)} d\theta \quad \because \frac{(m-1)(m-3)\cdots}{m(m-2)(m-4)\cdots}$$

$$= \int_0^{\pi/2} 64 \frac{(3)(1)}{4(2)} d\theta$$

$$= \int_0^{\pi/2} 64 \cdot \frac{3}{8} d\theta$$

$$= \int_0^{\pi/2} 24 \, d\theta \Rightarrow 24 \int_0^{\pi/2} d\theta$$

$$= 24 \, |\theta|_0^{\pi/2}$$

$$= 24 \left(\frac{\pi}{2} - 0\right)$$

$$= 24 \left(\frac{\pi}{2}\right)$$

$$= 12\pi \quad \text{Ans.}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Example:- Evaluate $\iint_R e^{x^2+y^2}\,dy\,dx$ where R is semicircular region bounded by x-axis & the curve $y=\sqrt{1-x^2}$

$$y=\sqrt{1-x^2}$$

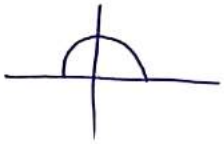
Take square on b/s

$$y^2=1-x^2$$

$$y^2+x^2=1 \quad \text{(comparing with } x^2+y^2=r^2\text{)}$$

here $r^2=1$, $r=1$ & $r=0$ (bcz semi circle

Now $\theta=0$ & $\theta=\pi$ (bcz semi-circle on x-axis)



$$=\frac{1}{2}\int_0^{\pi}\int_0^{1} e^{r^2}\,2r\,dr\,d\theta$$

$$=\frac{1}{2}\int_0^{\pi}\left|e^{r^2}\right|_0^1 d\theta \Rightarrow \frac{1}{2}\int_0^{\pi}(e^1-e^0)\,d\theta$$

$$=\frac{1}{2}\int_0^{\pi}(e-1)\,d\theta \Rightarrow \frac{1}{2}(e-1)\int_0^{\pi} d\theta$$

$$=\frac{1}{2}(e-1)(\theta)\Big|_0^{\pi} \Rightarrow \frac{1}{2}(e-1)(\pi-0)$$

$$=\frac{\pi}{2}(e-1)$$ Ans.

NOTES BY KINZA BILAL

Scanned with CamScanner

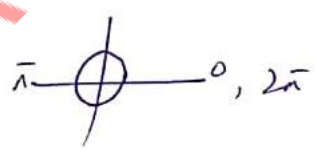
Example:- Let R_a be region bounded by the circle $x^2+y^2=a^2$. Define

$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)}\,dx\,dy = \lim_{a\to\infty}\iint_R e^{-(x^2+y^2)}\,dx\,dy$$

$x^2+y^2=a^2$ here $r^2=a^2$

$r=a$ & $r=0$

$\theta=0$ & $\theta=2\pi$ (bcz region is bounded by circle)



$$\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)}\,dx\,dy = \lim_{a\to\infty}\iint_R e^{-(x^2+y^2)}\,dx\,dy$$

$$= \lim_{a\to\infty} -\frac{1}{2}\int_0^{2\pi}\int_0^{a} e^{-r^2}(-2)r\,dr\,d\theta$$

$$= -\frac{1}{2}\lim_{a\to\infty}\int_0^{2\pi}\left|e^{-r^2}\right|_0^a d\theta$$

$$= -\frac{1}{2}\lim_{a\to\infty}\int_0^{2\pi}\left(e^{-a^2}-e^{-0}\right)d\theta$$

$$= -\frac{1}{2}\lim_{a\to\infty}\int_0^{2\pi}(e^{-a^2}-1)\,d\theta$$

$$= -\frac{1}{2}\lim_{a\to\infty}(e^{-a^2}-1)\int_0^{2\pi} d\theta$$

$$= -\frac{1}{2}\lim_{a\to\infty}(e^{-a^2}-1)|\theta|_0^{2\pi}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$= -\frac{1}{2} \lim_{a \to \infty} \left(e^{-a^2} - 1\right)\left(2\bar{\pi} - 0\right)$$

$$= \lim_{a \to \infty} \left(1 - e^{-a^2}\right)\left(\frac{2\bar{\pi}}{2}\right)$$

$$= \lim_{a \to \infty} \left(1 - \frac{1}{e^{a^2}}\right)\bar{\pi}$$

$$= \left(1 - \frac{1}{e^{\infty}}\right)\bar{\pi} \Rightarrow \left(1 - \frac{1}{\infty}\right)\bar{\pi}$$

$$= (1 - 0)\,\bar{\pi} \Rightarrow \bar{\pi}$$ Ans.

Example: Prove That $\int_0^{\infty} e^{-t^2}\,dt = \frac{1}{2}\sqrt{\pi}$

Can't solve directly bcz derivative of $-t^2$ is not given. So

let $I = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)}\,dx\,dy$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-x^2-y^2}\,dx\,dy$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-x^2}\cdot e^{-y^2}\,dx\,dy$$

$$= \left[\int_{-\infty}^{\infty} e^{-x^2}\,dx\right]\left[\int_{-\infty}^{\infty} e^{-y^2}\,dy\right]$$

NOTES BY KINZA BILAL

Scanned with CamScanner

use a dummy variable t.

$$= \left[\int_{-\infty}^{\infty} e^{-t^2}\,dt\right]\left[\int_{-\infty}^{\infty} e^{-t^2}\,dt\right]$$

$$= \left[\int_{-\infty}^{\infty} e^{-t^2}\,dt\right]^2$$

introduce a vawable a & apply limit

$$= \lim_{a\to\infty}\left(\int_{-a}^{a} e^{-t^2}\,dt\right)^2$$

$$= \lim_{a\to\infty} 4\left(\int_{0}^{a} e^{-t^2}\,dt\right)^2 \qquad \left(\int_0^a \text{ shows one quadrant}\right)$$

Here

$$4\lim_{a\to\infty}\left(\int_0^a e^{-t^2}\,dt\right)^2 = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-(x^2+y^2)}\,dx\,dy$$

$$4\lim_{a\to\infty}\left(\int_0^a e^{-t^2}\,dt\right)^2 = \pi \qquad \text{(reference to previous example).}$$

$$\left(\int_0^{\infty} e^{-t^2}\,dt\right)^2 = \frac{\pi}{4}$$

$$\int_0^{\infty} e^{-t^2}\,dt = \frac{\sqrt{\pi}}{2} \text{ or } \frac{1}{2}\sqrt{\pi}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Theorem:- Let G be a rectangular box defined by inequalities

$$a \le x \le b \;,\; c \le y \le d \;,\; k \le z \le l$$

If G is continuous then

$$\iiint_G f(x,y,z)\,dV = \int_a^b \int_c^d \int_k^l f(x,y,z)\,dz\,dy\,dx$$

order of integration doesn't matter.

Example:- Evaluate Tripple integral

$\iiint_G 12xy^2z^3\,dV$ over rectangular box G defined by inequalities

$$-1 \le x \le 2 \;,\; 0 \le y \le 3 \;,\; 0 \le z \le 2.$$

$$= \int_{-1}^{2}\int_{0}^{3}\int_{0}^{2} 12xy^2z^3\,dz\,dy\,dx$$

$$= \int_{-1}^{2}\int_{0}^{3} 12xy^2 \left|\frac{z^4}{4}\right|_0^2 dy\,dx$$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$= \int_{-1}^{2} \int_{0}^{3} \frac{12xy^2}{4} (2^4 - 0^4)\, dy\, dx$$

$$= \int_{-1}^{2} \int_{0}^{3} 3xy^2 (16)\, dy\, dx$$

$$= \int_{-1}^{2} \int_{0}^{3} 48xy^2\, dy\, dx$$

$$= \int_{-1}^{2} 48x \left| \frac{y^3}{3} \right|_0^3 dx$$

$$= \int_{-1}^{2} \frac{48x}{3} (3^3 - 0^3)\, dx$$

$$= \int_{-1}^{2} 16x (27)\, dx$$

$$= \int_{-1}^{2} 432x\, dx$$

$$= 432 \left| \frac{x^2}{2} \right|_{-1}^{2} dx$$

$$= \frac{432}{2} (2^2 - (-1)^2) \Rightarrow 216(4-1)$$

$$\Rightarrow 216(3) \Rightarrow 648 \text{ Ans.}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Example:- $\iiint_R (x-2y+z)\,dx\,dy\,dz$ when

$0 \le x \le 1$, $0 \le y \le x^2$, $0 \le z \le x+y$

$$\int_0^1 \int_0^{x^2} \int_0^{x+y} (x-2y+z)\,dz\,dy\,dx$$

$$= \int_0^1 \int_0^{x^2} \left| \frac{(x-2y+z)^2}{2} \right|_0^{x+y} dy\,dx$$

$$= \int_0^1 \int_0^{x^2} \left[\frac{(x-2y+x+y)^2}{2} - \frac{(x-2y)^2}{2} \right] dy\,dx$$

$$= \int_0^1 \int_0^{x^2} \left[\frac{(2x-y)^2 - (x-2y)^2}{2} \right] dy\,dx$$

$$= \int_0^1 \int_0^{x^2} \left(\frac{4x^2+y^2-4xy - x^2-4y^2+4xy}{2} \right) dy\,dx$$

$$= \frac{1}{2} \int_0^1 \int_0^{x^2} (3x^2-3y^2)\,dy\,dx$$

$$= \frac{3}{2} \int_0^1 \int_0^{x^2} (x^2-y^2)\,dy\,dx$$

$$= \frac{3}{2} \int_0^1 \left(x^2y - \frac{y^3}{3} \right)\Big|_0^{x^2} dx$$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$= \frac{3}{2}\int_0^1 \left(\left[x^2(x^2) - \frac{(x^2)^3}{3} \right] - \left[x^2(0) - \frac{0^3}{3} \right] \right) dx$$

$$= \frac{3}{2}\int_0^1 \left(x^4 - \frac{x^6}{3} \right) dx$$

$$= \frac{3}{2}\left(\frac{x^5}{5} - \frac{x^7}{3 \cdot 7} \right)\Big|_0^1$$

$$= \frac{3}{2}\left(\frac{x^5}{5} - \frac{x^7}{21} \right)\Big|_0^1$$

$$= \frac{3}{2}\left[\left(\frac{(1)^5}{5} - \frac{(1)^7}{21} \right) - \left(\frac{(0)^5}{5} - \frac{(0)^7}{21} \right) \right]$$

$$= \frac{3}{2}\left[\frac{1}{5} - \frac{1}{21} \right]$$

$$= \frac{3}{2}\left[\frac{21-5}{105} \right]$$

$$= \frac{3}{2}\left(\frac{16}{105} \right)$$

$$= \frac{8}{35} \quad \text{Ans.}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Example:- Evaluate $\iiint xyz \, dx\,dy\,dz$

where $S = \{(x,y,z) : x^2+y^2+z^2 \le 1,\ x \ge 0, y \ge 0, z \ge 0\}$

$x^2+y^2+z^2=1$

$z^2 = 1-x^2-y^2$

$z = \sqrt{1-x^2-y^2}$

ignore z

$x^2+y^2=1$

$y^2 = 1-x^2$

$y = \sqrt{1-x^2}$

ignore y.

$x^2=1$

$x = 1$

$$\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} xyz \, dz\,dy\,dx$$

$$= \int_0^1 \int_0^{\sqrt{1-x^2}} xy \left| \frac{z^2}{2} \right|_0^{\sqrt{1-x^2-y^2}} dy\,dx$$

$$= \int_0^1 \int_0^{\sqrt{1-x^2}} xy \frac{\left(\sqrt{1-x^2-y^2}\right)^2}{2} dy\,dx$$

$$= \frac{1}{2} \int_0^1 \int_0^{\sqrt{1-x^2}} xy(1-x^2-y^2) \, dy\,dx$$

$$= \frac{1}{2} \int_0^1 \int_0^{\sqrt{1-x^2}} x\left(y - x^2y - y^3\right) dy\,dx$$

$$= \frac{1}{2} \int_0^1 \left[x \left| \frac{y^2}{2} - \frac{x^2y^2}{2} - \frac{y^4}{4} \right|_0^{\sqrt{1-x^2}} \right] dx$$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$= \frac{1}{2}\int_0^1 x\left(\frac{(\sqrt{1-x^2})^2}{2} - \frac{x^2(\sqrt{1-x^2})^2}{2} - \frac{(\sqrt{1-x^2})^4}{4}\right) dx$$

$$= \frac{1}{2}\int_0^1 x\left(\frac{1-x^2}{2} - \frac{x^2(1-x^2)}{2} - \frac{(1-x^2)^2}{4}\right) dx$$

$$= \frac{1}{2}\int_0^1 x\left(\frac{1-x^2}{2} - \frac{x^2-x^4}{2} - \frac{1+x^4-2x^2}{4}\right) dx$$

$$= \frac{1}{2}\int_0^1 x\left(\frac{2-2x^2-2x^2+2x^4-1-x^4+2x^2}{4}\right) dx$$

$$= \frac{1}{2}\int_0^1 x\left(\frac{1-2x^2+x^4}{4}\right) dx$$

$$= \frac{1}{8}\int_0^1 \frac{x-2x^3+x^5}{} dx$$

$$= \frac{1}{8}\int_0^1 x^5 - 2x^3 + x \quad dx$$

$$= \frac{1}{8}\left[\frac{x^6}{6} - \frac{2x^4}{4} + \frac{x^2}{2}\right]_0^1$$

$$= \frac{1}{8}\left[\frac{(1)^6}{6} - \frac{2(1)^4}{4} + \frac{(1)^2}{2}\right] - 0$$

$$= \frac{1}{8}\left(\frac{1}{6} - \frac{2}{4} + \frac{1}{2}\right) \Rightarrow \frac{1}{8}\left(\frac{1-3+3}{6}\right) \Rightarrow \frac{1}{12}$$ Ans.

NOTES BY KINZA BILAL

Scanned with CamScanner

Lecture 27

Vector Valued Functions

Those functions for which the domain consists of real numbers & the range consists of vectors in 2d-space & or 3d-space are called Vector valued functions.

Such functions can be expressed in

2d-space like $r(t) = (x(t), y(t))$

$= x(t)i + y(t)j$

Such functions can be expressed in

3d-space like $r(t) = (x(t), y(t), z(t))$

$= x(t)i + y(t)j + z(t)k$

here $x(t)$, $y(t)$ & $z(t)$ are called real valued functions of the real variable t. These functions are called component functions or components of r.

NOTES BY KINZA BILAL

Scanned with CamScanner

Representation:-

Vector valued functions are represented by bold letters $\mathbf{f(t)}$, $\mathbf{g(t)}$ etc

Real valued functions are represented by italic lightface type. $f(t)$, $g(t)$ etc.

Parametric Equations in vector Form:-

Vector valued functions can be used to express parametric equations in 2-space/3-space in a compact Form.

For example consider parametric equations

$$x = x(t) \quad \& \quad y = y(t).$$

Two vectors are equivalent iff their corresponding components are equal, so this pair of equations can be replaced by single vector equation.

NOTES BY KINZA BILAL

Scanned with CamScanner

$x = x(t)$, $y = y(t)$

$x\underline{i} + y\underline{j} = x(t)\underline{i} + y(t)\underline{j}$

Similarly for Three dimensional space These equations

$x = x(t)$, $y = y(t)$, $z = z(t)$

can be replaced by single vector equation

$$x\underline{i} + y\underline{j} + z\underline{k} = x(t)\underline{i} + y(t)\underline{j} + z(t)\underline{k}$$

In 2-d

$r = x\underline{i} + y\underline{j}$ & $r(t) = x(t)\underline{i} + y(t)\underline{j}$ are same.

In 3-d

$r = x\underline{i} + y\underline{j} + z\underline{k}$ & $r(t) = x(t)\underline{i} + y(t)\underline{j} + z(t)\underline{k}$

are same.

So we can write $r = r(t)$.

Example:- Express given parametric equ. in single vector equ.

(a) $x = t^2$ & $y = 3t$

$$r = x\underline{i} + y\underline{j} = t^2\underline{i} + 3t\underline{j}$$

(b) $x = \cos t$, $y = \sin t$, $z = t$

$$r = x\underline{i} + y\underline{j} + z\underline{k} = \cos t\,\underline{i} + \sin t\,\underline{j} + t\,\underline{k}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Example:- Find parametric eq. That correspond to the vector equation.

$$x\underline{i} + y\underline{j} + z\underline{k} = (t^3+1)\underline{i} + 3\underline{j} + e^t\underline{k}$$

Comparing

$$x = t^3+1 \;,\quad y = 3 \;,\quad z = e^t$$

Example:- Sketch graph of vector-valued function $r(t) = \cos t\,\underline{i} + \sin t\,\underline{j} \quad 0 \le t \le 2\pi$

As $r(t) = x\underline{i} + y\underline{j}$

$\cos t\,\underline{i} + \sin t\,\underline{j} = x\underline{i} + y\underline{j}$

Comparing

$$x = \cos t \;\&\; y = \sin t \quad (0 \le t \le 2\pi)$$

Take square on b/s

$$x^2 = \cos^2 t \;\&\; y^2 = \sin^2 t$$

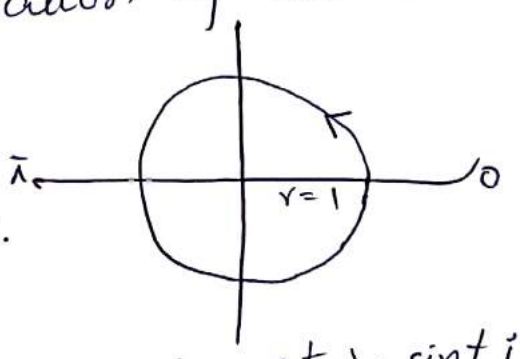
Now adding both equations

$$x^2 + y^2 = \cos^2 t + \sin^2 t$$

$$x^2 + y^2 = 1$$

So we got an equation of circle whose radius is 1.

in anti-clockwise direction.



$r = \cos t\,\underline{i} + \sin t\,\underline{j}$

NOTES BY KINZA BILAL

Scanned with CamScanner

Example:- Sketch the graph of vector-valued function $r(t) = \cos t\, i + \sin t\, j + 2k \quad 0 \le t \le 2\pi$

As $r = r(t)$

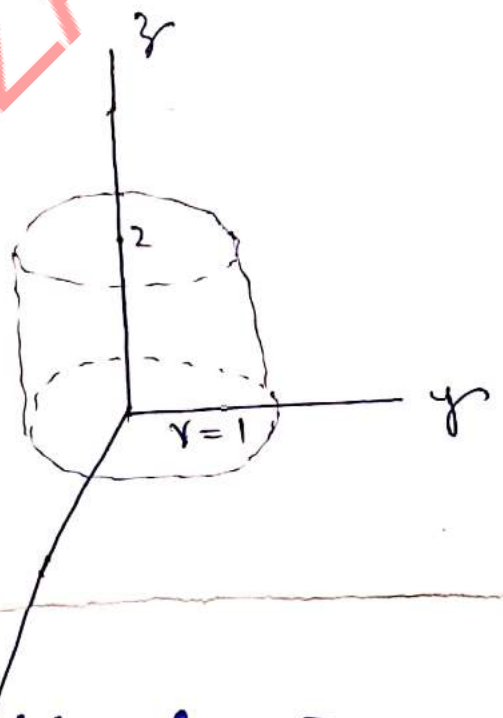
$r = xi + yj + zk$

$\cos t\, i + \sin t\, j + 2k = xi + yj + zk$

Comparing $x = \cos t$, $y = \sin t$, $z = 2$

squaring & adding first two equations

$x^2 + y^2 = 1$



stretch along z-axis.

$r = \cos t\, i + \sin t\, j + 2k$

Example:- Describe the graph of vector equ.

$r = (-2+t)i + 3tj + (5-4t)k$

Comparing $x = -2+t$, $y = 3t$, $z = 5-4t$

Graph is a line passing through $(-2, 0, 5)$ & parallel to $i + 3j - 4k$

(ignoring constants & t).

NOTES BY KINZA BILAL

Scanned with CamScanner

Example:- The graph of vector-valued func $r(t) = t\underline{i} + t^2\underline{j} + t^3\underline{k}$ is called twisted cubic. Show That This curve ~~lies~~ Lies on parabolic cyllinder $y = x^2$ & sketch graph for $t \geq 0$.

$$r(t) = t\underline{i} + t^2\underline{j} + t^3\underline{k}$$

Corresponding equations are:-

$$x = t, \quad y = t^2, \quad z = t^3$$

If we put $x = t$ in $y = t^2$ it becomes $y = x^2$ which is equation of parabolic cyllinder. So This curve Lies on parabolic cyllinder.

If we start t from origion ($t = 0$) by increasing "t" values of x, y, z also increases.

Graphs of constant vector-valued func.

If c is a constant vector in The sense That it does not depend on parameter, Then The graph of $r = c$ is a single point since The radius vector remains fixed with its tip at c.

NOTES BY KINZA BILAL

Scanned with CamScanner

In 2d-space If $c = x_0 \underline{i} + y_0 \underline{j}$, then graph is the point (x_0, y_0) & if $c = x_0 \underline{i} + y_0 \underline{j} + z_0 k$ (3d-space) then graph is point (x_0, y_0, z_0).

x_0, y_0, z_0 are constant components.

Example:- The graph of $r(t) = \cos t \underline{i} + \sin t \underline{j} + 2k$

As $x = \cos t$ & $y = \sin t$ & $z = 2$

They form equ. of circle with radius 1 & lying in plane $z = 2$.

$$|r(t)| = \sqrt{x^2 + y^2 + z^2} = \sqrt{\cos^2 t + \sin^2 t + (2)^2}$$

$$= \sqrt{1 + 4} = \sqrt{5}$$

So the circle lies on surface of a sphere of radius $\sqrt{5}$.

NOTES BY KINZA BILAL

Scanned with CamScanner

Lecture 28

Limits of vector valued functions

Limit of vector valued function is defined to be vector that results by taking limit of each component.

In 2d-Space $r(t) = x(t)\underline{i} + y(t)\underline{j}$ is defined as

$$\lim_{t\to\alpha} r(t) = \lim_{t\to\alpha} x(t)\underline{i} + \lim_{t\to\alpha} y(t)\underline{j}$$

In 3d-space $r(t) = x(t)\underline{i} + y(t)\underline{j} + z(t)\underline{k}$ is defined as

$$\lim_{t\to\alpha} r(t) = \lim_{t\to\alpha} x(t)\underline{i} + \lim_{t\to\alpha} y(t)\underline{j} + \lim_{t\to\alpha} z(t)\underline{k}$$

If lim of any component does not exist Then limit of $r(t)$ does not exist.

Same goes for $\lim_{t\to\alpha^+}$, $\lim_{t\to\alpha^-}$, $\lim_{t\to\infty^+}$, $\lim_{t\to\infty^-}$

We can write $\lim_{t\to\alpha} r(t) = L$

Length & direction of $r(t)$ approaches to length & direction of L as $t\to\alpha$.

NOTES BY KINZA BILAL

Scanned with CamScanner

Continuity of vector valued functions

r is continuous at t_0 if

$r(t_0)$ is defined.

$\lim_{t \to t_0} r(t)$ exists.

$\lim_{t \to t_0} r(t) = r(t_0)$

The graph of a continuous vector-valued function is unbroken curve.

Derivatives of vector valued functions:-

The vector-valued function's derivative is

$$r'(t) = \lim_{h \to 0} \frac{r(t+h) - r(t)}{h}$$

$\frac{f(x+h) - f(x)}{h}$

Theorem:- If $r(t) = x(t)\underline{i} + y(t)\underline{j}$ in 2d.

Then $r'(t) = x'(t)\underline{i} + y'(t)\underline{j}$

Similarly if $r(t) = x(t)\underline{i} + y(t)\underline{j} + z(t)\underline{k}$ in 3d Then $r'(t) = x'(t)\underline{i} + y'(t)\underline{j} + z'(t)k$

Proof:-

As $r'(t) = \lim_{h \to 0} \frac{r(t+h) - r(t)}{h}$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$r'(t) = \lim_{h\to 0} \frac{[x(t+h)-x(t)]}{h}\underline{i} + \lim_{h\to 0}\frac{[y(t+h)-y(t)]}{h}\underline{j}$$

$x(t)\underline{i} + y(t)\underline{j}$

$$= x'(t)\underline{i} + y'(t)\underline{j}$$

Similarly for 3d-space.

Notations:-

$$\frac{d}{dt}r(t), \quad \frac{dr}{dt}, \quad r'(t), \quad \dot{r}$$

Example:- $r(t) = t^2\underline{i} + t^3\underline{j}$, $r'(t) = ?$

$r'(1) = ?$

$$r'(t) = \frac{d}{dt}(t^2)\underline{i} + \frac{d}{dt}(t^3)\underline{j}$$

$$r'(t) = 2t\underline{i} + 3t^2\underline{j}$$

now put $t = 1$

$$r'(1) = 2(1)\underline{i} + 3(1)^2\underline{j}$$

$$r'(1) = 2\underline{i} + 3\underline{j}$$

Tangent vectors & Tangent lines

geometrical interpretation of derivative

Let P be a point on graph of vector-valued func $r(t)$ & let $r(t_0)$ be radius vector from origin to P. If $r'(t_0)$ exists & $r'(t_0) \neq 0$ then we call

NOTES BY KINZA BILAL

Scanned with CamScanner

$r'(t_0)$ The tangent vector to the graph of r at $r(t_0)$

Equation:- Vector equation of Tangent line is

$$r = r(t_0) + t\, r'(t_0)$$

Example:- Find parametric equ. of Tangent line to circular helix $x=\cos t$, $y=\sin t$, $z=t$ at point where $t_0 = \frac{\pi}{6}$.

Equating $r(t) = x\underline{i} + y\underline{j} + z\underline{k}$

$$r(t) = (\cos t)\underline{i} + (\sin t)\underline{j} + (t)k \longrightarrow (1)$$

$$r'(t) = (-\sin t)\underline{i} + (\cos t)\underline{j} + k \longrightarrow (2)$$

Put $t = \frac{\pi}{6}$ in (1) & (2) respectively

$$r\left(\frac{\pi}{6}\right) = \left(\cos\frac{\pi}{6}\right)\underline{i} + \left(\sin\frac{\pi}{6}\right)\underline{j} + \left(\frac{\pi}{6}\right)\underline{k}$$

$$r\left(\frac{\pi}{6}\right) = \frac{\sqrt{3}}{2}\,\underline{i} + \frac{1}{2}\,\underline{j} + \frac{\pi}{6}k \quad \&$$

$$r'\left(\frac{\pi}{6}\right) = -\frac{1}{2}\,\underline{i} + \frac{\sqrt{3}}{2}\,\underline{j} + k$$

Now equ. of Tangent

$$r = r(t_0) + t\, r'(t_0)$$

$$= \left(\frac{\sqrt{3}}{2}\,\underline{i} + \frac{1}{2}\,\underline{j} + \frac{\pi}{6}\underline{k}\right) + t\left(\frac{-1}{2}\,\underline{i} + \frac{\sqrt{3}}{2}\,\underline{j} + k\right)$$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$= \left(\frac{\sqrt{3}}{2}\underline{i} + \frac{1}{2}\underline{j} + \frac{\pi}{6}\underline{k}\right) + -\frac{1}{2}t\,\underline{i} + \frac{\sqrt{3}}{2}t\,\underline{j} + tk$$

$$= \left(\frac{\sqrt{3}}{2} - \frac{1}{2}t\right)\underline{i} + \left(\frac{1}{2} + \frac{\sqrt{3}}{2}t\right)\underline{j} + \left(\frac{\pi}{6} + t\right)\underline{k}$$

Comparing with $r = x\underline{i} + y\underline{j} + z\underline{k}$

$$x = \frac{\sqrt{3}}{2} - \frac{1}{2}t \;,\quad y = \frac{1}{2} + \frac{\sqrt{3}}{2}t \;,\quad z = \frac{\pi}{6} + t$$

Example:- The graph of $r(t) = t^2\underline{i} + t^3\underline{j}$ is called a semi-cubical parabola. Find a vector equation of tangent line to graph of $r(t)$ at

(a) The point $(0,0)$

(b) The point $(1,1)$

$$r(t) = t^2\underline{i} + t^3\underline{j}$$

$$r'(t) = 2t\underline{i} + 3t^2\underline{j}$$

(a) here $t=0$ so $r'(0) = 0$. There is no Tangent vector at the point & Tangent Line does not exist.

NOTES BY KINZA BILAL

Scanned with CamScanner

(b) here $t=1$ So vector equation will be

$$r = r(t_0) + t\, r'(t_0)$$

$$r = r(1) + t\, r'(1) \quad \text{—— (1)}$$

$$r(t) = t^2 \underline{i} + t^3 \underline{j}$$

So $r(1) = \underline{i} + \underline{j}$

& $r'(t) = 2t\underline{i} + 3t\underline{j}$

$r'(1) = 2\underline{i} + 3\underline{j}$

putting values in (1)

$$r = r(1) + t r'(1)$$

$$= (\underline{i} + \underline{j}) + t(2\underline{i} + 3\underline{j})$$

$$= \underline{i} + \underline{j} + 2t\underline{i} + 3t\underline{j}$$

$$= (1+2t)\underline{i} + (1+3t)\underline{j}$$

Smooth Function:-

$r(t)$ is called smooth function if components of r have continuous derivatives w.r.t t.

& $r'(t) \neq 0$ for any value of t.

Same for 2d & 3d- space.

A parametric eq will be called smooth if has a Tangent line at every point on its graph.

NOTES BY KINZA BILAL

Scanned with CamScanner

Properties of derivatives:-

- $\frac{d}{dt}(c) = 0$
- $\frac{d}{dt}\left[k\, r(t)\right] = k\,\frac{d}{dt}\left[r(t)\right]$
- $\frac{d}{dt}\left[r_1(t) + r_2(t)\right] = \frac{d}{dt}\left[r_1(t)\right] + \frac{d}{dt}\left[r_2(t)\right]$
- $\frac{d}{dt}\left[r_1(t) - r_2(t)\right] = \frac{d}{dt}\left[r_1(t)\right] - \frac{d}{dt}\left[r_2(t)\right]$
- $\frac{d}{dt}\left[r_1(t) \cdot r_2(t)\right] = \left[r_1(t)\right]\frac{d}{dx}\left[r_2(t)\right] + \left[r_2(t)\right]\frac{d}{dx}\left[r_1(t)\right]$

Dot Product in 2d.

- $\frac{d}{dt}\left[r_1(t) \cdot r_2(t)\right] = r_1 \cdot \frac{dr_2}{dt} + \frac{dr_1}{dt} \cdot r_2$

Cross product in 3d.

- $\frac{d}{dt}\left[r_1(t) \times r_2(t)\right] = r_1 \times \frac{dr_2}{dt} + \frac{dr_1}{dt} \times r_2$

Order doesn't matter in 2d (Dot Product)
But matters in 3d (Cross Product)

Theorem:- If $r(t)$ is a vector-valued function in 2 space or 3 space & $|r(t)|$ is constant for all t, then $r(t) \cdot r'(t) = 0$ (orthogonal vectors)

NOTES BY KINZA BILAL

Scanned with CamScanner

keeping $r_1(t) = r_2(t) = r(t)$ derivative of Dot product becomes

$$\frac{d}{dt}\left[r(t).r(t)\right] = r(t).\frac{dr}{dt} + \frac{dr}{dt}.r(t)$$

or

$$\frac{d}{dt}\left[|r(t)|\right]^2 = 2r(t).\frac{dr}{dt}$$

But $\left[|r(t)|\right]^2$ is constant, so its derivative is zero. So

$$2r(t)\frac{dr}{dt} = 0$$

$$r(t).\frac{dr}{dt} = 0$$

(i-e) $r(t)$ is perpendicular to $\frac{dr}{dt}$.

Integrals of vector valued functions:-

(a) if $r(t) = x(t)\underline{i} + y(t)\underline{j}$, then

$$\int r(t)\,dt = \left(\int x(t)\,dt\right)\underline{i} + \left(\int y(t)\,dt\right)\underline{j}$$

$$\int_a^b r(t)\,dt = \left(\int_a^b x(t)\,dt\right)\underline{i} + \left(\int_a^b y(t)\,dt\right)\underline{j}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

If $r(t) = x(t)\underline{i} + y(t)\underline{j} + z(t)\underline{k}$

then

$$\int r(t)\,dt = \left(\int x(t)\,dt\right)\underline{i} + \left(\int y(t)\,dt\right)\underline{j} + \left(\int z(t)\,dt\right)\underline{k}$$

$$\int_a^b r(t)\,dt = \left(\int_a^b x(t)\,dt\right)\underline{i} + \left(\int_a^b y(t)\,dt\right)\underline{j} + \left(\int_a^b z(t)\,dt\right)\underline{k}$$

Example: $r(t) = 2t\,\underline{i} + 3t^2\,\underline{j}$

(a) $\int r(t)\,dt$

$= \int (2t\,\underline{i} + 3t^2\,\underline{j})\,dt$

$= \left(\int 2t\,dt\right)\underline{i} + \left(\int 3t^2\,dt\right)\underline{j}$

$= \left(\frac{2t^2}{2} + C_1\right)\underline{i} + \left(\frac{3t^3}{3} + C_2\right)\underline{j}$

$= t^2\underline{i} + C_1\underline{i} + t^3\underline{j} + C_2\underline{j}$

$= t^2\underline{i} + t^3\underline{j} + C$ where $C_1\underline{i} + C_2\underline{j} = C$

(b) $\int_0^2 r(t)\,dt$

$= \left[t^2\right]_0^2\underline{i} + \left[t^3\right]_0^2\underline{j}$

$= (2^2 - 0^2)\underline{i} + (2^3 - 0^3)\underline{j}$

$= 4\underline{i} + 8\underline{j}$

NOTES BY KINZA BILAL

Scanned with CamScanner

Properties of integral:-

$\frac{d}{dx}\left(\int x^2 dx\right)$

$$\int c\, r(t)\, dt = c \int r(t)\, dt$$

$$\int [r_1(t) + r_2(t)]\, dt = \int r_1(t)\, dt + \int r_2(t)\, dt$$

$$\int [r_1(t) - r_2(t)]\, dt = \int r_1(t)\, dt - \int r_2(t)\, dt$$

These properties also hold for definite integral.

In addition $\frac{d}{dt}\left(\int r(t)\, dt\right) = r(t)$

This shows indefinite integral of $r(t)$'s set of antiderivatives of $r(t)$ just as for real valued function.

$$\int r(t)\, dt = R(t) + C$$

$$\int_a^b r(t)\, dt = R(t)\Big]_a^b = R(b) - R(a).$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Lecture 29.

Change of Parameter

It is possible for different valued functions to have same graph. For example

$$r = (3\cos t)\hat{i} + 3(\sin t)\hat{j} \qquad 0 \le t \le 2\pi$$

&

$$r = \left[3\cos\left(\frac{s}{3}\right)\right]\hat{i} + \left[3\sin\left(\frac{s}{3}\right)\right]\hat{j} \qquad 0 \le s \le 6\pi$$

both the function have same graph ↑ circle of radius centered at origin. (here we made substitution $t = \frac{s}{3}$. real valued function). We can say that $t = g(u)$.

New vector valued function of u is smooth if original vector valued function of t is smooth.

If $\Rightarrow g$ is differentiable.

$\Rightarrow g'$ is continuous.

$\Rightarrow g'(u) \neq 0$ for any u in domain of g.

NOTES BY KINZA BILAL

Scanned with CamScanner

Arc length:-

⇒ If $x = x(t)$ & $y(t)$ in 2-space

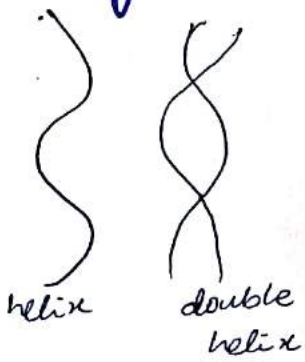
Then arc length $L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$

⇒ If $x = x(t)$, $y = y(t)$ & $z = z(t)$ is 3-space

arc length $L = \int_a^b \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 + \left(\frac{dz}{dt}\right)^2}\, dt$

Example:- Find arc length of that portion of the circular helix $x = \cos t$, $y = \sin t$, $z = t$

Take $t = 0$ & $t = \pi$, $\frac{dx}{dt} = -\sin t$, $\frac{dy}{dt} = \cos t$, $\frac{dz}{dt} = 1$



$$L = \int_0^{\pi} \sqrt{(-\sin t)^2 + (\cos t)^2 + (1)^2}\, dt$$

$$= \int_0^{\pi} \sqrt{\sin^2 t + \cos^2 t + 1}\, dt$$

$$= \int_0^{\pi} \sqrt{2}\, dt$$

$$= \sqrt{2} \int_0^{\pi} 1\, dt$$

$$= \sqrt{2}\,(t)\Big]_0^{\pi} \Rightarrow \sqrt{2}\,(\pi - 0)$$

$$L = \sqrt{2}\,\pi \quad \text{Ans.}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Arc length as a parameter:-

The best parameter to use for representing a curve in 2-space or 3-space parametrically is length of arc measured along the curve from some reference point.

Steps:-

⇒ Select an arbitrary point on curve as reference point. (P)

⇒ Starting from reference point, choose one direction along curve ~~as~~ to be +ve & other to be -ve.

⇒ Let s be an arc Then s is +ve if point is +ve & ~~conversely~~ vice versa.

Theorem:- (a) Let C be a curve in 2-space parametrically by $x = x(t)$ & $y = y(t)$ where $x'(t)$ & $y'(t)$ are continuous functions. if S is arc length Then

$$S = \int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}\, du$$

(b) Let C be a curve in 3-space parametrically by $x = x(t)$, $y = y(t)$ & $z = z(t)$ where $x'(t)$, $y'(t)$ & $z'(t)$ are continuous, Then arc length

NOTES BY KINZA BILAL

Scanned with CamScanner

$$S = \int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2 + \left(\frac{dz}{du}\right)^2}$$

Proof:- if $t > t_0$

$$S = \int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}\, du$$

$$\frac{ds}{dt} = \frac{d}{dt}\left[\int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}\, dy\right.$$

$$= \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

Similarly for 3-d.

$\frac{ds}{dt}$ does not depend on t_0, its value is independent.

Example:- $x = 2t + 1$, $y = 3t - 2$

here reference point $(1, -2)$

replace t by u

$$x = 2u + 1, \quad y = 3u - 2$$

$$\frac{dx}{du} = 2, \quad \frac{dy}{du} = 3$$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$S = \int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}\, du$$

$$= \int_{t_0}^{t} \sqrt{(2)^2 + (3)^2}\, du$$

$$= \int_{t_0}^{t} \sqrt{13}\, du$$

$$= \sqrt{13} \int_{0}^{t} 1\, du \Rightarrow \sqrt{13}\, [u]_0^t$$

$$= \sqrt{13}(t-0) \Rightarrow \sqrt{13}\, t$$

So $s = \sqrt{13}\, t$

$$\frac{s}{\sqrt{13}} = t$$

put in

$$x = 2t + 1 \Rightarrow 2\left(\frac{s}{\sqrt{13}}\right) + 1 \Rightarrow \frac{2}{\sqrt{13}} s + 1$$

$$y = 3(t) - 2 \Rightarrow 3\left(\frac{s}{\sqrt{13}}\right) - 2 \Rightarrow \frac{3}{\sqrt{13}} s - 2$$

So These are parametric equations.

NOTES BY KINZA BILAL

Scanned with CamScanner

Example:- Find parametric equations for the circle $x = a\cos t$, $y = a\sin t$ using arc length s as parameter, with reference point for s being $(a, 0)$, $a > 0$. $\quad 0 \le t \le 2\pi$

Replace t by u.

$$x = a\cos u, \quad y = a\sin u$$

$$\frac{dx}{du} = -a\sin u, \quad \frac{dy}{du} = a\cos u$$

Now
$$s = \int_{t_0}^{t} \sqrt{\left(\frac{dx}{du}\right)^2 + \left(\frac{dy}{du}\right)^2}\, du$$

$$= \int_{t_0}^{t} \sqrt{(-a\sin u)^2 + (a\cos u)^2}\, du$$

$$= \int_{0}^{t} \sqrt{a^2\sin^2 u + a^2\cos^2 u}\, du$$

$$= \int_{0}^{t} \sqrt{a^2(\sin^2 u + \cos^2 u)}\, du$$

$$= \int_{0}^{t} \sqrt{a^2}\, du \Rightarrow \int_{0}^{t} a\, du \Rightarrow a\int_{0}^{t} du$$

$$= a[u]_0^t \Rightarrow a(t - 0) \Rightarrow at$$ Ans.

$$s = at, \text{ & } t = \frac{s}{a}, \text{ So}$$

$$x = a\cos\left(\frac{s}{a}\right) \text{ & } y = a\sin\left(\frac{s}{a}\right)$$

$0 \le t \le 2\pi$
$0 \le \frac{s}{a} \le 2\pi$
$0 \le s \le 2\pi a$

NOTES BY KINZA BILAL

Scanned with CamScanner

Example:- Find arc length of curve

$$r(t) = t^3 \underline{i} + t \underline{j} + \frac{1}{2}\sqrt{6}\, t^2 \underline{k} \qquad 1 \le t \le 3$$

$$x = t^3, \quad y = t, \quad z = \frac{1}{2}\sqrt{6}\, t^2$$

$$\frac{dx}{dt} = 3t^2, \quad \frac{dy}{dt} = 1, \quad \frac{dz}{dt} = \frac{1}{2}\sqrt{6}(2t) = \sqrt{6}\, t$$

$$\text{Arc length} = \int_1^3 \sqrt{(3t^2)^2 + (1)^2 + (\sqrt{6}\, t)^2}\, dt$$

$$= \int_1^3 \sqrt{9t^4 + 1 + 6t^2}\, dt$$

$$= \int_1^3 \sqrt{(3t^2)^2 + (1)^2 + 2(3t^2)(1)}\, dt$$

$$= \int_1^3 \sqrt{(3t^2+1)^2}\, dt$$

$$= \int_1^3 (3t^2+1)\, dt$$

$$= \left[\frac{3t^3}{3} + t\right]_1^3 \Rightarrow \left[t^3 + t\right]_1^3$$

$$= (3^3 + 3) - (1^3 + 1) \Rightarrow (27 + 3) - (1+1)$$

$$= 30 - 2 \Rightarrow 28 \quad \underline{\text{Ans}}.$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Lecture 30

Exact Differential

If $z = f(x,y)$, Then $dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy$

If $z = f(x,y,w)$, Then $dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy + \frac{\partial z}{\partial w}dw$

Exercise

1. $z = x^2 + y^2$, $dz = 2x\,dx + 2y\,dy \Rightarrow 2(x\,dx + y\,dy)$

2. $z = x^3 \sin 2y$, $dz = 3x^2 \cdot \sin 2y\,dx + x^3 \cos 2y \cdot 2\,dy$

$= x^2(3\sin 2y\,dx + 2x\cos 2y\,dy)$

3. $z = (2x-1)e^{3y}$, $dz = 2e^{3y}dx + (2x-1)e^{3y}\cdot 3dy$

$= e^{3y}(2dx + 3(2x-1)dy)$

4. $z = x^2 + 2y^2 + 3w^2$, $dz = 2x\,dx + 4y\,dy + 6w\,dw$

$= 2(x\,dx + 2y\,dy + 3w\,dw)$

5. $z = x^3y^2w$, $dz = 3x^2y^2w\,dx + x^3(2y)w\,dy + x^3y^2\,dw$

$= x^2y(3ywdx + 2xw\,dy + xy\,dw)$

NOTES BY KINZA BILAL

Scanned with CamScanner

Exact differential:-

As in $z = f(x,y)$, $dz = \frac{\partial z}{\partial x}dx + \frac{\partial z}{\partial y}dy$

Work in reverse now

Any expression like $dz = Pdx + Qdy$ can be written as $P = \frac{\partial z}{\partial x}$ & $Q = \frac{\partial z}{\partial y}$

Now $\frac{\partial P}{\partial y} = \frac{\partial^2 z}{\partial y \partial x}$ & $\frac{\partial Q}{\partial x} = \frac{\partial^2 z}{\partial x \partial y}$

Also $\frac{\partial^2 z}{\partial y \partial x} = \frac{\partial^2 z}{\partial x \partial y}$

So dz is exact differential if $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

Example:- $dz = (3x^2+4y^2)dx + 8xy\,dy$

Comparing with $Pdx + Qdy$

$P = 3x^2+4y^2$ & $Q = 8xy$

$\frac{\partial P}{\partial y} = 8y$ $\qquad$ $\frac{\partial Q}{\partial x} = 8y$

So dz is an exact differential.

NOTES BY KINZA BILAL

Scanned with CamScanner

Example:- $dz = (1+8xy)\,dx + 5x^2\,dy$

here $P = 1+8xy$ & $Q = 5x^2$

$\frac{\partial P}{\partial y} = 8x$ $\qquad$ $\frac{\partial Q}{\partial x} = 10x$

Both are not equal so dz is not an exat differential.

Exercise:-

1. $dz = 4x^3y^3\,dx + 3x^4y^2\,dy$

$P = 4x^3y^3$ | $Q = 3x^4y^2$

$\frac{\partial P}{\partial y} = 4x^3(3y^2)$ | $\frac{\partial Q}{\partial x} = 3(4x^3)y^2$

$= 12x^3y^2$ | $= 12x^3y^2$

dz is exact differential.

2. $dz = (4x^3y + 2xy^3)\,dx + (x^4 + 3x^2y^2)\,dy$

$P = 4x^3y + 2xy^3$ | $Q = x^4 + 3x^2y^2$

$\frac{\partial P}{\partial y} = 4x^3(1) + 2x(3y^2)$ | $\frac{\partial Q}{\partial x} = 4x^3 + 3(2x)y^2$

$= 4x^3 + 6xy^2$ | $= 4x^3 + 6xy^2$

YES

NOTES BY KINZA BILAL

Scanned with CamScanner

3. $dz = (15y^2e^{3x} + 2xy^2)dx + (10ye^{3x} + x^2y)dy$

$P = 15y^2e^{3x} + 2xy^2$

$\frac{\partial P}{\partial y} = 15(2y)e^{3x} + 2x(2y)$

$= \underline{30ye^{3x} + 4xy}$

$Q = 10ye^{3x} + x^2y$

$\frac{\partial Q}{\partial x} = 10ye^{3x} \cdot 3 + 2xy$

$= \underline{30ye^{3x} + 2xy}$

dz is not exact differential.

4. $dz = (3x^2e^{2y} - 2y^2e^{3x})dx + (2x^3e^{2y} - 2ye^{3x})dy$

$P = 3x^2e^{2y} - 2y^2e^{3x}$

$\frac{\partial P}{\partial y} = 3x^2e^{2y} \cdot 2 - 4y^2e^{3x}$

$= \underline{6x^2e^{2y} - 4y^2e^{3x}}$

NO

$Q = 2x^3e^{2y} - 2ye^{3x}$

$\frac{\partial Q}{\partial x} = 6xe^{2y} - 2ye^{3x} \cdot 3$

$= \underline{6x^2e^{2y} - 6ye^{3x}}$

5. $dz = (4y^3\cos 4x + 3x^2\cos 2y)dx + (3y^2\sin 4x - 2x^3\sin 2y)dy$

$P = 4y^3\cos 4x + 3x^2\cos 2y$

$\frac{\partial P}{\partial y} = 4(3y^2)\cos 4x + 3x^2(-\sin 2y) \cdot 2$

$= \underline{12y^2\cos 4x - 6x^2\sin 2y}$

$Q = 3y^2\sin 4x - 2x^3\sin 2y$

$\frac{\partial Q}{\partial x} = 3y^2(\cos 4x) \cdot 4 - 2(3x^2)\sin 2y$

$= \underline{12y^2\cos 4x - 6x^2\sin 2y}$

YES

NOTES BY KINZA BILAL

Scanned with CamScanner

Integration of Exact differentials

$dz = P\,dx + Q\,dy$ where $P = \frac{\partial z}{\partial x}$ & $Q = \frac{\partial z}{\partial y}$

$\therefore z = \int P\,dx$ & $z = \int Q\,dy$

Example:- $dz = (2xy + 6x)\,dx + (x^2 + 2y^3)\,dy$

$P = 2xy + 6x$, $Q = x^2 + 2y^3$

$z = \int P\,dx$

$= \int (2xy + 6x)\,dx$

$= 2y \cdot \frac{x^2}{2} + 6\frac{x^2}{2} + F(y)$

$z = x^2y + 3x^2 + F(y)$

$z = \int Q\,dy$

$= \int (x^2 + 2y^3)\,dy$

$= x^2y + 2\frac{y^4}{4} + F(x)$

$z = x^2y + \frac{y^4}{2} + F(x)$

To make them equal $F(y)$ should be $\frac{y^4}{2}$ & $F(x)$ should be $3x^2$

So $z = x^2y + 3x^2 + \frac{y^4}{2}$

Example: Integrate $dz = (8e^{4x} + 2xy^2)\,dx + (4\cos 4y + 2x^2y)\,dy$. to make

$z = 2e^{4x} + x^2y^2 + \sin 4y$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$dz = (8e^{4x} + 2xy^2)dx + (4\cos 4y + 2x^2y)dy$$

here

$$P = 8e^{4x} + 2xy^2$$

$$z = \int P\,dx$$

$$= \int (8e^{4x} + 2xy^2)\,dx$$

$$= \frac{8e^{4x}}{4} + 2\frac{x^2}{2}\cdot y^2 + F(y)$$

$$= 2e^{4x} + x^2y^2 + F(y)$$

$$Q = 4\cos 4y + 2x^2y$$

$$z = \int Q\,dy$$

$$z = \int (4\cos 4y + 2x^2y)\,dy$$

$$= \frac{4(\sin 4y)}{4} + 2x^2\frac{y^2}{2} + f(x)$$

$$= \sin 4y + x^2y^2 + F(x)$$

here $F(y) = \sin 4y$ & $F(x) = 2e^{4x}$

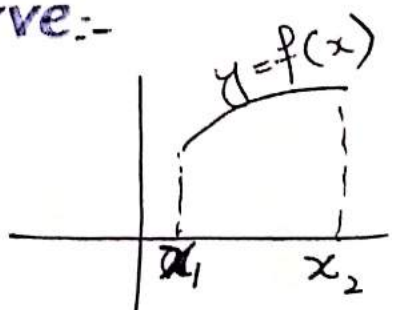
So

$$z = 2e^{4x} + x^2y^2 + \sin 4y.$$

Area enclosed by closed curve:-

Remember we used integration to find area of a plane.

$$A_1 = \int_{x_1}^{x_2} y\,dx = \int_{x_1}^{x_2} f(x)\,dx$$

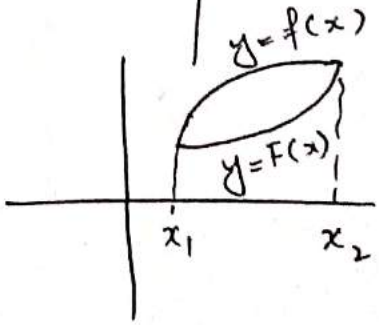


If There are Two curves

$$A = A_1 - A_2$$

$$A = \int_{x_1}^{x_2} f(x)\,dx - \int_{x_1}^{x_2} F(x)\,dx$$

$$= -\left\{ \int_{x_1}^{x_2} f(x)\,dx + \int_{x_1}^{x_2} F(x)\,dx \right.$$



NOTES BY KINZA BILAL

Scanned with CamScanner

Final result can also be written as

$$A = -\oint y\,dy$$

$\oint$ indicates that integral is to be ~~written~~ evaluated round closed boundry in the +ve.

Example:- Determine area enclosed by

The graph of $y = x^3$ & $y = 4x$ for $x \geq 0$.

$$Y = x^3 \text{ \& } y = 4x$$

To find limits

$$x^3 = 4x$$
$$x^3 - 4x = 0$$
$$x(x^2 - 4) = 0$$

$x = 0$

$x^2 - 4 = 0$
$x^2 = 4$
$\sqrt{x^2} = \sqrt{(2)^2}$
$x = \pm 2$
$x = 2 \quad \because x \geq 0$

in anticlockwise direction

C_1 : $y = x^3$, limit $x = 0$ & $x = 2$

C_2 : $y = 4x$, limit $x = 2$ & $x = 0$

$$A = -\oint y\,dy \Rightarrow -\left\{ \int_0^2 x^3\,dx + \int_2^0 4x\,dx \right\}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$= -\left\{ \left[\frac{x^4}{4}\right]_0^2 + \left[4\frac{x^2}{2}\right]_2^0 \right\}$$

$$= -\left(\left[\frac{(2)^4}{4} - \frac{(0)^4}{2}\right] + \left[2(0)^2 - 2(2)^2\right] \right)$$

$$= -\left(\frac{16}{4} - 8\right)$$

$$= -(4-8) \Rightarrow -(-4) \Rightarrow 4 \text{ Ans.}$$

Example:- Find area of Triangle with vertices $O(0,0)$ $A(5,3)$ & $B(2,6)$.

m = Slope of $OA = \frac{y_2 - y_1}{x_2 - x_1} = \frac{3-0}{5-0} = \frac{3}{5}$

m = Slope of $BA = \frac{6-3}{2-5} = \frac{3}{-3} = -1$

m = Slope of $OB = \frac{6-0}{2-0} = \frac{6}{2} = 3$

eq. of $OA \Rightarrow y - y_1 = m(x - x_1)$

$$y - 0 = \frac{3}{5}(x-0)$$

$$y = \frac{3}{5}x$$

eq. of $BA \Rightarrow y - 5 = -1(x-3)$

$$y = -x + 3 + 5$$

$$y = -x + 8$$

NOTES BY KINZA BILAL

Scanned with CamScanner

eq. of OB ⇒ $y-0 = 3(x-0)$

$y = 3x$

Now

$$A = -\oint y\,dy$$

$$= -\left(\int_0^5 \frac{3}{5}x\,dx + \int_5^2 (-x+8)\,dy + \int_2^0 3x\,dx \right)$$

$$= -\left(\frac{3}{5}\left[\frac{x^2}{2}\right]_0^5 + \left[\frac{-x^2}{2} + 8x\right]_5^2 + \left[\frac{3x^2}{2}\right]_2^0 \right)$$

$$= -\left(\frac{3}{5}\left[\frac{5^2-0^2}{2}\right] + \left[\left(\frac{-(2)^2}{2} + 8(2)\right) - \left(\frac{(5)^2}{2} + 8(5)\right)\right] + \left[\quad \right] \right.$$

$$\left. \left[\frac{3(0)^2}{2} - \frac{3(2)^2}{2}\right] \right)$$

$$= -\left[\frac{3}{5}\left(\frac{25}{2}\right) + \left(14 - \frac{55}{2}\right) + (0-6)\right]$$

$$= -\left(\frac{15}{2} + 14 - \frac{55}{2} - 6\right)$$

$$= -\left(\frac{15+28-55-12}{2}\right) \Rightarrow -\left(\frac{-24}{2}\right)$$

$$\Rightarrow 12 \text{ Ans.}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Line integral:-

If a field exists in xy-plane, producing a Force F on a particle at k. Then F can be divided into two components.

F_1 along Tangent & F_a along normal.

Approximate workdone in moving particle is $F_t \delta s$ if δs is the distance covered.

$$W = \sum F_t \delta s$$

$$= \lim_{\delta s \to 0} \sum F_t \delta x = \int F_t \, ds$$

or $$\int_{AB} F_t \, ds = \int_C F_t \, ds$$

where C is curve $y = f(x)$ between $A(x_1, y_1)$ & $B(x_2, y_2)$.

Alternative form:-

If F_t has a component P in x-direction & Q in y-direction then workdone is

$$P\delta x + Q\delta y$$

NOTES BY KINZA BILAL

Scanned with CamScanner

So $\int_{AB} F_t \, ds = \int_{AB} (P\,dx + Q\,dy)$

P & Q are functions of x & y.

In general.

$$\int_C F_t \, ds = \int_C (P\,dx + Q\,dy) \quad \left(\text{where } c \text{ is the curve}\right)$$

Lecture 31 Line Integral.

Example:- Evaluate $\int_C (x+3y)\,dx$ from $A(0,1)$ to $B(2,5)$ along the curve $y = 1+x^2$

In $\int_C (x+3y)\,dx$, $P = x+3y$ & $Q = 0$ & $c = 1+x^2$

$$\int_C (x+3(1+x^2))\,dx = \int_0^2 (x+3+3x^2)\,dx$$

$$= \left[\frac{x^2}{2} + 3x + \frac{3x^3}{3}\right]_0^2 = \left[\frac{2^2}{2} + 3(2) + (2)^3\right] - 0$$

$$= (2+6+8) \Rightarrow 16 \quad \text{Ans.}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Example 2:- Evaluate $I = \int_C (x^2+y)dx + (x-y^2)dy$ from $A(0,2)$ to $B(2,5)$ along curve $y = 2+x$

$y = 2+x$

$dy = dx$

$P = x^2+y \Rightarrow x^2+2+x$

$Q = x-y^2 \Rightarrow x-(2+x)^2 \Rightarrow x-4-x^2-4x$

$= -3x-4-x^2 \Rightarrow -(3x+4+x^2)$

So

$$\int_C (P\,dx + Q\,dy)$$

$$= \int_0^2 \left[(x^2+x+2)\frac{dx}{} + (-(3x+4+x^2))\,dx \right]$$

$$= \int_0^2 \left[x^2+x+2-3x-4-x^2 \right] dx$$

$$= \int_0^2 (-2x-2)\,dx \Rightarrow \left[-\frac{2x^2}{2} - 2x \right]_0^2$$

$$= \left(-(2)^2 - 2(2) \right) - \left(-0^2 - 2(0) \right)$$

$$= (-4-4) - 0$$

$$= -8 \quad \text{Ans.}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Example Evaluate $I = \int_C (x^2+2y)dx + xy\,dy$ from $(0,0)$ To $B(1,4)$ along curve $y = 4x^2$

$y = 4x^2$

$dy = 8x\,dx$

$P = x^2+2y \Rightarrow x^2+2(4x^2)$

$= x^2+8x^2 \Rightarrow 9x^2$

$Q = xy \Rightarrow x(4x^2) \Rightarrow 4x^3$

So

$$\int_C P\,dx + Q\,dy$$

$$= \int_0^1 9x^2\,dx + 4x^3(8x\,dx)$$

$$= \int_0^1 (9x^2 + 32x^4)\,dx \Rightarrow \left[\frac{9x^3}{3} + 32\frac{x^5}{5}\right]_0^1$$

$$\Rightarrow \left(\frac{9(1)^3}{3} + 32\frac{(1)^5}{5}\right) - \left(\frac{9(0)^3}{3} + 32\frac{(0)^5}{5}\right)$$

$$= \left(\frac{9}{3} + \frac{32}{5}\right) - 0 \Rightarrow 3 + \frac{32}{5} \Rightarrow \frac{15+32}{5}$$

$$= \frac{47}{5} \Rightarrow 9.4 \quad \text{Ans.}$$

Example:- Evaluate $\int_C (x^2+2y)dx + xy\,dy$ from $O(0,0)$ to $A(1,0)$ along line $y=0$ & Then from $A(1,0)$ To $B(1,4)$ along line $x=1$.

NOTES BY KINZA BILAL

Scanned with CamScanner

(i) for line $y=0$ & $O(0,0)$, $A(1,0)$

$P = x^2+2y \Rightarrow x^2+2(0) = x^2$

$Q = xy \Rightarrow x(0) \Rightarrow 0$

$y = 0$
$dy = 0$

So

$$I_{OA} = \int_C P\,dx + Q\,dy$$

$$= \int_0^1 x^2\,dx + 0 \Rightarrow \int_0^1 x^2\,dx \Rightarrow \left[\frac{x^3}{3}\right]_0^1$$

$$= \left(\frac{(1)^3}{3} - \frac{0^3}{3}\right) \Rightarrow \frac{1}{3}$$

(ii) for line $x=1$ & $A(1,0)$ To $B(1,4)$

$P = x^2+2y \Rightarrow 1^2+2y \Rightarrow 1+2y$

$Q = xy = (1)y \Rightarrow y$

$x = 1$
$dx = 0$

So

$$I_{AB} = \int_C P\,dx + Q\,dy \Rightarrow \int_0^4 (1+2y)(0)+(y)\,dy$$

$$= \int_0^4 y\,dy \Rightarrow \left[\frac{y^2}{2}\right]_0^4 \Rightarrow \frac{(4)^2}{2} \Rightarrow \frac{16}{2} \Rightarrow 8$$

Then $I = I_{OA} + I_{AB}$

$$= \frac{1}{3} + 8 \Rightarrow \frac{1+24}{3} \Rightarrow \frac{25}{3} \text{ or } 8\frac{1}{3} \text{ Ans.}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Properties of line integral:-

$\Rightarrow \int_C F\,ds = \int_c (P\,dx + Q\,dy)$

$\Rightarrow \int_{AB} F\,ds = -\int_{BA} F\,ds$

or

$\int_{AB} (P\,dx + Q\,dy) = -\int_{BA} (P\,dx + Q\,dy).$

Sign is reversed when direction is reversed.

$\Rightarrow$ For a path parallel to x-axis i.e $x = k$, $dx = 0$

So $\int_C P\,dx = 0 \quad \therefore \int_C Q\,dy$

$\Rightarrow$ For a path parallel to y-axis i.e $y = k$, $dy = 0$

$\int_C Q\,dy = 0 \quad \therefore \int_C P\,dx$

$\Rightarrow$ If path is not single valued, then it is divided into two sections. $y = f_1(x)$ A to K & $y = f_2(x)$ from K to B.

$\Rightarrow$ If path $\bar{AB}$ is divided into two ~~paths~~ parts AK & BK then $I_c = I_{AB} = I_{AK} + I_{KB}$

NOTES BY KINZA BILAL

Scanned with CamScanner

$\Rightarrow$ In all cases actual path must be continuous & single valued.

Example:- Evaluate $I = \int_C (x+y)\,dx$ from $A(0,1)$ to $B(0,-1)$ along semi-circle $x^2+y^2=1$ for $x \geq 0$

here path is not single valued

$$x^2+y^2=1, \quad y^2 = 1-x^2$$

$$y = \pm\sqrt{1-x^2}$$

So (i) $y = \sqrt{1-x^2}$ from A to K

(ii) $y = -\sqrt{1-x^2}$ from K to B.

As $I = \int_C (x+y)\,dx$ here $Q = 0$

$P = x+y \Rightarrow x+\sqrt{1-x^2}$

So $I = \int_C (P\,dx + Q\,dy)$

will be $I = \int_C P\,dx$

$$I_{AB} = I_{AK} + I_{KB}$$

$$= \int_0^1 (x+\sqrt{1-x^2})\,dx + \int_1^0 (x-\sqrt{1-x^2})\,dx$$

$$= \int_0^1 (x+\sqrt{1-x^2})\,dx - \int_0^1 (x-\sqrt{1-x^2})\,dx$$

$$= \int_0^1 (x+\sqrt{1-x^2} - x + \sqrt{1-x^2})\,dx$$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$= \int_0^1 2\sqrt{1-x^2}\,dx$$

$$= 2\int_0^1 \sqrt{1-x^2}\,dx$$

If we put $x = \sin\theta$

$dx = \cos\theta\, d\theta$

put $x = 0$ in $x = \sin\theta$

$0 = \sin\theta$

$\theta = 0$

put $x = 1$

$1 = \sin\theta$

$\sin^{-1}(1) = \theta$

$\frac{\pi}{2} = \theta$

$$= 2\int_0^{\pi/2} \sqrt{1-\sin^2\theta}\,\cos\theta\,d\theta$$

$$= 2\int_0^{\pi/2} \cos\theta\cdot\cos\theta\,d\theta$$

$$= 2\int_0^{\pi/2} \cos^2\theta\,d\theta$$

$$= 2\int_0^{\pi/2} \left(\frac{1+\cos 2\theta}{2}\right) d\theta$$

$\because \cos^2\frac{\theta}{2} = \frac{1+\cos\theta}{2}$

$$= \int_0^{\pi/2} (1+\cos 2\theta)\,d\theta \Rightarrow$$

$$= \left[\theta + \frac{\sin 2\theta}{2}\right]_0^{\frac{\pi}{2}}$$

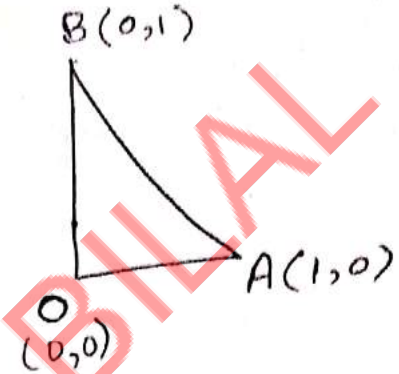
$$= \left(\frac{\pi}{2} + \frac{\sin\pi}{2}\right) - \left(0 - \frac{\sin 0}{2}\right)$$

$$= \left(\frac{\pi}{2} + 0\right) - 0 \Rightarrow \frac{\pi}{2}$$ Ans.

NOTES BY KINZA BILAL

Scanned with CamScanner

Example:- Evaluate $\oint (x^2 dx - 2xy\,dy)$ where c comprises three sides of triangle joining O(0,0) A(1,0) & B(0,1).



For OA, $y = 0$

$dy = 0$

$P = x^2$, $Q = 2xy \Rightarrow 2x(0) \Rightarrow 0$

So

$$I_1 = \int_{OA} x^2 dx \Rightarrow \int_0^1 x^2 dx \Rightarrow \left| \frac{x^3}{3} \right|_0^1$$

$$I_1 = \frac{(1)^3}{3} - \frac{(0)^3}{3} \Rightarrow \frac{1}{3} - 0 \Rightarrow \boxed{\frac{1}{3}}$$

For AB, $y = 1 - x$

$dy = -1\,dx$

$P = x^2$, $Q = -2xy$

$P = x^2$, $Q = -2x(1-x) \Rightarrow -2x + 2x^2$

So

$$I_2 = \int_{AB} x^2 dx + (-2x + 2x^2) - dx$$

$$\int_1^0 (x^2 + 2x - 2x^2)\, dx$$

$y - y_0 = m(x - x_0)$

$y - 0 = -1(x - 1)$

$y = -x + 1$

$y = 1 - x$

$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{0-1}{1-0} = -1$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$I_2 = \int_1^0 (-x^2 + 2x)\,dx$$

$$= \left[-\frac{x^3}{3} + \frac{2x^2}{2} \right]_1^0 \Rightarrow 0 - \left(-\frac{1^3}{3} + (1)^2 \right)$$

$$I_2 = -\left(-\frac{1}{3} + 1 \right) \Rightarrow -\left(\frac{-1+3}{3} \right) \Rightarrow \boxed{-\frac{2}{3}}$$

For BO, line $x = 0$

$dx = 0$

$P = x^2$, $Q = -2xy$

$P = 0$, $Q = 0$

$$I_3 \int_{BO} 0\,dy \Rightarrow 0$$

Finally $I = I_1 + I_2 + I_3$

$$= \frac{1}{3} - \frac{2}{3} + 0$$

$$= \frac{1-2}{3} \Rightarrow \boxed{\frac{-1}{3}} \text{ Ans.}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Example:- Evaluate $\oint_c y\,dx$ when c is the circle $x^2+y^2=4$.

$$y^2 = 4 - x^2$$

$$y = \pm\sqrt{4-x^2}$$

y is not single valued so

$x^2 = 4$

$x = x = \pm 2$

$$I = \int_2^{-2} \sqrt{4-x^2}\,dx + \int_{-2}^{2} -\sqrt{4-x^2}\,dx$$

$$= \int_{+2}^{-2} \sqrt{4-x^2}\,dx + \int_{2}^{-2} +\sqrt{4-x^2}\,dx$$

$$= \int_{2}^{-2} 2\sqrt{4-x^2}\,dx$$

$$= 2\int_{-2}^{2} \sqrt{4-x^2}\,dx$$

$$= 2\times 2\int_{0}^{2} \sqrt{4-x^2}\,dx$$

$$= 4\int_{0}^{2} \sqrt{4-x^2}\,dx$$

NOTES BY KINZA BILAL

Scanned with CamScanner

put $x = 2\sin\theta$,

$dx = 2\cos\theta\, d\theta$

Limits	
$x = 0$	$x = 2$
$x = 2\sin\theta$	$2 = 2\sin\theta$
$0 = 2\sin\theta$	$1 = \sin\theta$
$0 = \sin\theta$	$\sin^{-1}(1) = \theta$
$\sin^{-1}(0) = \theta$	$\frac{\pi}{2} = \theta$
$0 = \theta$	

Now

$$-4\int_0^{\pi/2} \sqrt{4-(2\sin\theta)^2}\, 2\cos\theta\, d\theta$$

$$= -4\int_0^{\pi/2} \sqrt{4-4\sin^2\theta}\, 2\cos\theta\, d\theta$$

$$= -4\int_0^{\pi/2} \sqrt{4(1-\sin^2\theta)}\, 2\cos\theta\, d\theta$$

$$= -4\int_0^{\pi/2} 2\cos\theta\, 2\cos\theta\, d\theta$$

$$= -4\times 2\int_0^{\pi/2} 2\cos^2\theta\, d\theta \qquad \because 2\cos^2\frac{\theta}{2} = 1+\cos\theta$$

$$= -16\int_0^{\pi/2} \left(\frac{1+\cos 2\theta}{2}\right) d\theta$$

$$= -8\int_0^{\pi/2} (1+\cos 2\theta)\, d\theta \Rightarrow -8\left[\theta + \frac{\sin 2\theta}{2}\right]_0^{\frac{\pi}{2}}$$

$$= -8\left[\frac{\pi}{2} + \frac{\sin 2\left(\frac{\pi}{2}\right)}{2}\right] \Rightarrow -8\left[\frac{\pi}{2}+0\right] \Rightarrow -8\left(\frac{\pi}{2}\right)$$

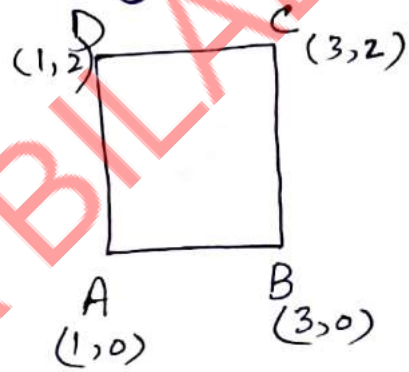
$$= -4\pi \quad \text{Ans.}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Lecture 32. Examples

Evaluate $I = \oint \{xy\,dx + (1+y^2)\,dy\}$ where c is boundry of rectangle joining A(1,0), B(3,0), C(3,2), D(1,2).



I_1 = line $y=0$, $dy=0$

$I_{AB} = I_1 = 0$

$I_2 = I_{BC}$ line $x=3$, $dx=0$

$$I_2 = \int_0^2 (1+y^2)\,dy \Rightarrow \left[y + \frac{y^3}{3}\right]_0^2 \Rightarrow \left(2 + \frac{2^3}{3}\right) - 0$$

$$= 2 + \frac{8}{3} \Rightarrow \frac{6+8}{3} = \frac{14}{3}$$

$I_3 = I_{CD}$ $y=2$, $dy=0$

$$I_3 = \int_3^1 x(2)\,dx \Rightarrow \left[2\,\frac{x^2}{2}\right]_3^1 = \left[x^2\right]_3^1$$

$$= (1)^2 - (3)^2 \Rightarrow 1 - 9 \Rightarrow -8$$

$I_4 = I_{DA}$ line $x=1$, $dx=0$

$$I_4 = \int_2^0 (1+y^2)\,dy \Rightarrow \left[y + \frac{y^3}{3}\right]_2^0 \Rightarrow 0 - \left(2 + \frac{2^3}{3}\right)$$

$$\Rightarrow \frac{-14}{3}$$ Ans

NOTES BY KINZA BILAL

Scanned with CamScanner

Now $I = I_1 + I_2 + I_3 + I_4$

$= 0 + \frac{14}{3} - 8 - \frac{14}{3} \Rightarrow -8$ Ans.

Line integral w.r.t arc length

As $I = \int_{AB} F_t \, ds = \int_{AB} (P dx + Q dy)$

if $I = \int_c f(x,y)\, ds = \int_c f(x,y) \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx$

where $\frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$

Example:- Evaluate $I = \int_c (4x + 3xy)\, ds$ where c is straight line joining O(0,0) to A(1,2). c is line $y = 2x$

$dy = 2$

$\frac{ds}{dx} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2}$

$= \sqrt{1 + (2)^2} \Rightarrow \sqrt{5}$

$y - y_0 = m(x - x_0)$

$y - 0 = 2(x - 0)$

$y = 2x$

$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{2-0}{1-0} = 2$

Now

$\int_0^1 (4x + 3xy)\sqrt{5}\, dx$

$\int_0^1 (4x + 3x(2x))\sqrt{5}\, dx$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$\int_0^1 (4x + 6x^2)\sqrt{5}\, dx$$

$$= \sqrt{5}\left[\frac{4x^2}{2} + \frac{6x^3}{3}\right]_0^1$$

$$= \sqrt{5}\left[2x^2 + 2x^3\right]_0^1$$

$$= \sqrt{5} \times 2\left[(1^2 + 1^3) - (0)\right]$$

$$= 2\sqrt{5}\,(2) \Rightarrow 4\sqrt{5} \text{ Ans.}$$

Parametric Equations:-

When x & y are expressed in parametric form $x = f(t)$ & $y = g(t)$, Then

$$\frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}$$

$$I = \int_c f(x,y)\, ds = \int_{t_1}^{t_2} f(x,y) \sqrt{\left(\frac{dx}{dt}\right)^2 + \frac{dy}{(dt)^2}}\, dt$$

Example:- Evaluate $I = \oint 4xy\, ds$, where c is defined as curve $x = \sin t$, $y = \cos t$. between $t = 0$ & $t = \frac{\pi}{4}$.

NOTES BY KINZA BILAL

Scanned with CamScanner

$$x = \sin t \;,\; y = \cos t$$

$$\frac{dx}{dt} = \cos t \;,\; \frac{dy}{dt} = -\sin t$$

$$\frac{ds}{dt} = \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2} \Rightarrow \sqrt{\cos^2 t + (-\sin t)^2}$$

$$= \sqrt{1} = 1$$

$$I = \int_{t_1}^{t_2} f(x,y) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

$$= \int_0^{\pi/4} 4xy\,(1)\,dt \Rightarrow \int_0^{\pi/4} 4 \sin t \cos t\, dt$$

$$= 2\int_0^{\pi/4} 2\sin t \cos t\, dt \Rightarrow 2\int_0^{\pi/4} \sin 2t\, dt$$

$$= +2\left[-\frac{\cos 2t}{2}\right]_0^{\frac{\pi}{4}} \Rightarrow -\left[\cos 2t\right]_0^{\frac{\pi}{4}}$$

$$= -\left[\cos 2\left(\frac{\pi}{4}\right) - \cos 2(0)\right]$$

$$= -\left(\cos\frac{\pi}{2} - \cos 0\right)$$

$$= -(0 - 1) \Rightarrow -(-1) \Rightarrow 1 \quad \underline{\text{Ans.}}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Dependence of linear integral on path of integration.

Example:- Evaluate $I = \oint \{3x^2y^2dx + 2x^3y\,dy\}$

b/w $O(0,0)$ & $A(2,4)$

(a) along C_1 (i-e) $y = x^2$

(b) along C_2 (ie) $y = 2x$

(c) along C_3 (i-e) $x = 0$ from $(0,0)$ to $(0,4)$

& $y = 4$ from $(0,4)$ to $(2,4)$

(a)

$$I_1 = \int_C (3x^2y^2dx + 2x^3y\,dy) \quad \text{line} \quad y = x^2, \; dy = 2x\,dx$$

$$= \int_0^2 \left(3x^6dx + 2x^3(x^2)(2x)\,dx\right)$$

$$= \int_0^2 (3x^6 + 4x^6)\,dx \Rightarrow \int_0^2 7x^6\,dx$$

$$= 7\left[\frac{x^7}{7}\right]_0^2 \Rightarrow [x^7]_0^2 \Rightarrow (2)^7 - (0)^7 \Rightarrow 128$$

$$\boxed{I_1 = 128}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

(b) line $y = 2x$

$dy = 2dx$

$$I_2 = \int_C (3x^2y^2dx + 2x^3y\,dy)$$

$$= \int_0^2 \left[3x^2(2x)^2dx + 2x^3(2x)(2dx) \right]$$

$$= \int_0^2 \left[12x^4 + 8x^4 \right]dx \Rightarrow \int_0^2 20x^4\,dx$$

$$= 20\left[\frac{x^5}{5}\right]_0^2 \Rightarrow 4[x^5]_0^2 \Rightarrow 4[2^5 - 0^5]$$

$$= 4(32) \Rightarrow 128$$

(c) $I_{3a} = 0$

line $x = 0$ from (0,0) to (0,4)

$dx = 0$

$$I_{3b} = \int_C 3x^2(4)^2dx$$

line $y = 4$ from (0,4) to (2,4)

$dy = 0$

$$= \int_0^2 48x^2dx$$

$$= 48\left[\frac{x^3}{3}\right]_0^2 \Rightarrow 16[x^3]_0^2 \Rightarrow 16[2^3 - 0^3]$$

$$= 16(8) \Rightarrow 128 \quad \underline{\text{Ans.}}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Lecture 33 Examples

Evaluate $I = \int_c \{3y\,dx + (3x+2y)dy\}$ from $A(1,2)$ to $B(3,5)$.

If no path is given, then integrand is an exact differential of some function $z = f(x,y)$. As

$$I = \int_c \{P\,dx + Q\,dy\}$$

here $P = 3y$, $z = \int 3y\,dx = 3xy + f(y)$

$Q = 3x+2y$, $z = \int (3x+2y)dy = 3xy + y^2 + f(x)$

in this case $f(y) = y^2$ & $f(x) = 0$

hence $z = 3xy + y^2$

$$I = \int_{(1,2)}^{(3,5)} d(3xy+y^2) \Rightarrow \left[3xy+y^2\right]_{(1,2)}^{(3,5)}$$

$$= \left[3(3)(5)+(5)^2\right] - \left[3(1)(2)+(2)^2\right]$$

$$= 3\left[45+25\right] - \left[6+4\right] \Rightarrow 70 - 10$$

$$= 60 \quad \text{Ans.}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Example:-

Evaluate $I = \int_C \{(x^2+ye^x)dx + (e^x+y)dy\}$ b/w

A(0,1) & B(1,2).

$P = x^2+ye^x$, $z = \int (x^2+ye^x)dx$

$= \frac{x^3}{3} + ye^x + f(y)$

$Q = (e^x+y)$, $z = \int Q\,dy \Rightarrow \int (e^x+y)dy$

$= e^x y + \frac{y^2}{2} + f(x)$

here $f(y) = \frac{y^2}{2}$ & $f(x) = \frac{x^3}{3}$

So $z = \frac{x^3}{3} + ye^x + \frac{y^2}{2}$

Then $I = \left[\frac{x^3}{3} + ye^x + \frac{y^2}{2}\right]_{(0,1)}^{(1,2)}$

$= \left(\frac{1}{3} + (2)e^1 + \frac{2^2}{2}\right) - \left(0 + (1)e^0 + \frac{1^2}{2}\right)$

$= \left(\frac{1}{3} + 2e + 2\right) - \left(1 + \frac{1}{2}\right)$

$= \frac{1+6e+6}{3} - \frac{3}{2} \Rightarrow \frac{12e+14-9}{6}$

$= \frac{12e+5}{6} \Rightarrow 2e + \frac{5}{6}$ Ans.

NOTES BY KINZA BILAL

Scanned with CamScanner

If $(Pdx + Qdy)$ is exact ~~function~~ differential

(a) $I = \int_c (Pdx + Qdy)$ is independent of path.

(b) $I = \oint_c (Pdx + Qdy)$ is zero.

Exact differentials in 3 independent variables:-

If $z = f(x, y, w)$ Then its exact differentia will be $dz = Pdx + Qdy + R\,dw$

& $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$; $\frac{\partial P}{\partial w} = \frac{\partial R}{\partial x}$; $\frac{\partial R}{\partial y} = \frac{\partial Q}{\partial w}$

if test is successful Then

(a) $\int_c (Pdx + Qdy + R\,dw)$ is independent of path.

(b) $\oint_c (Pdx + Qdy + Rdw)$ is zero.

Example:- Verify that

$dz = (3x^2yw + 6x)dx + (x^3w - 8y)dy + (x^3y + 1)dw$

is in exact differential & hence evaluate

$\int_c dz$ from $A(1,2,4)$ to $B(2,1,3)$.

NOTES BY KINZA BILAL

Scanned with CamScanner

$P = 3x^2yw + 6x$, $Q = x^3w - 8y$, $R = x^3y + 1$

$\frac{\partial P}{\partial y} = 3x^2w$, $\frac{\partial Q}{\partial x} = 3x^2w$, $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

$\frac{\partial P}{\partial w} = 3x^2y$, $\frac{\partial R}{\partial x} = 3x^2y$, $\frac{\partial P}{\partial w} = \frac{\partial R}{\partial x}$

$\frac{\partial Q}{\partial w} = x^3$, $\frac{\partial R}{\partial y} = x^3$, $\frac{\partial Q}{\partial w} = \frac{\partial R}{\partial y}$

So dz is exact differential.

To find z,

$z = \int P\,dx = \int (3x^2yw + 6x)\,dx \Rightarrow \frac{3x^3}{3}yw + \frac{6x^2}{2}$

$= x^3yw + 3x^2 + f(y) + F(w)$

$z = \int Q\,dy = \int (x^3w - 8y)\,dy \Rightarrow x^3yw - \frac{8y^2}{2}$

$= x^3yw - 4y^2 + F(w) + g(x)$

$z = \int R\,dy = \int (x^3y + 1)\,dw \Rightarrow x^3yw + w + f(y) + g(x)$

Now

$z = x^3yw + 3x^2 - 4y^2 + w$

$I = [x^3yw + 3x^2 - 4y^2 + w]$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$I = \left[x^3yw + 3x^2 - 4y^2 + w\right]_{(1,2,4)}^{(2,1,3)}$$

$$= \left[(2)^3(1)(3) + 3(2)^2 - 4(1)^2 + 3\right] - \left[(1)^3(2)(4) + 3(1)^2 - 4(2)^2 + 4\right]$$

$$= (24 + 12 - 4 + 3) - (8 + 3 - 16 + 4)$$

$$= 36 \quad \underline{\text{Ans.}}$$

Green's Theorem:- Let P & Q are two functions of x & y that are finite & continuous. If first partial derivatives are continuous, then

$$\iint_R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) dx\,dy = -\oint_C (P\,dx + Q\,dy)$$

Example:- Evaluate $I = \oint_C \{(2x - y)dx + (2y + x)dy\}$ around boundry C, the ellipse $x^2 + 9y^2 = 16$.

$P = 2x - y, \quad Q = 2y + x$

$\frac{\partial P}{\partial y} = 0 - 1 \Rightarrow -1, \quad \frac{\partial P}{\partial x} = 0 + 1 \Rightarrow 1$

$$I = -\iint_R (-1 - 1)\,dx\,dy = -\iint_R (-2)\,dx\,dy$$

$$= 2\iint_R dx\,dy$$ (Find area of ellipse & multip by 2 now)

NOTES BY KINZA BILAL

Scanned with CamScanner

$x^2 + 9y^2 = 16$

divide by 16

$\frac{x^2}{16} + \frac{9y^2}{16} = 1$ comparing with $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

$\frac{x^2}{(4)^2} + \frac{y^2}{\frac{(4)^2}{(3)^2}} = 1$

$a = 4, \quad b = \frac{4}{3}$

Area of ellipse $= \pi(a.b)$

$= \pi\left(4 \times \frac{4}{3}\right) \Rightarrow \frac{16}{3}\pi$

Now put area in

$= 2\iint_R dx\,dy \Rightarrow 2A$

$= 2\left(\frac{16}{3}\pi\right) \Rightarrow \frac{32}{3}\pi$ Ans.

Example:- Evaluate $I = \oint_C [(2x+y)dx + (3x-2y)dy]$ taken in anticlockwise around the triangle with vertices O(0,0), A(1,0), B(1,

$I = \oint_C [(2x+y)dx + (3x-2y)dy]$

$P = 2x+y, \quad \frac{\partial P}{\partial y} = 1$

$Q = 3x-2y, \quad \frac{\partial Q}{\partial x} = 3$

(not equal so apply green's Theorem)

NOTES BY KINZA BILAL

Scanned with CamScanner

$$I = -\iint_R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) dx\,dy$$

$$= -\iint_R (1 - 3)\, dx\,dy$$

$$= -\iint_R (-2)\, dx\,dy$$

$$= 2\iint_R dx\,dy \Rightarrow 2A$$

area of Triangle $= \frac{1}{2}$ (base x height)

$= \frac{1}{2}(1 \times 2)$

$= 1$

base = x value of A

height = y value of B.

put in

$2A = 2(1) = 2$ Ans.

NOTES BY KINZA BILAL

Scanned with CamScanner

Lecture 34 Examples

Examples:- Evaluate the line integral

$I = \oint_C \{xy\,dx + (2x-y)\,dy\}$ round the region bounded by the curves $y = x^2$ & $\sqrt{x} = \sqrt{y}$ by use of Green's Theorem.

$$I = \oint_C \{xy\,dx + (2x-y)\,dy\}$$

Take $y = x^2$ & $x = y^2$

$x = (x^2)^2$

$x = x^4$

$x - x^4 = 0$

$x(1-x^3) = 0$

$\boxed{x = 0}$ | $1 - x^3 = 0$

$1 = x^3$

$\boxed{1 = x}$

put $x = 0$ in $y = x^2$

$\boxed{y = 0}$

$(0,0)$

put $x = 1$ in $y = x^2$

$y = (1)^2$

$\boxed{y = 1}$

$(1,1)$

points of intersection

Now $P = xy$ & $Q = 2x - y$

$\frac{\partial P}{\partial y} = x(1) \Rightarrow x$ & $\frac{\partial Q}{\partial x} = 2(1) - 0 \Rightarrow 2$

Both are not equal so use green's Theorem.

NOTES BY KINZA BILAL

Scanned with CamScanner

$$I = \oint_C (P\,dx + Q\,dy) = -\iint_R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) dx\,dy$$

$$= -\int_0^1 \int_{x^2}^{\sqrt{x}} (x-2)\,dy\,dx$$

$$= -\int_0^1 (x-2)\left[y\right]_{x^2}^{\sqrt{x}}\,dx$$

$$= -\int_0^1 (x-2)\left(\sqrt{x} - x^2\right) dx$$

$$= -\int_0^1 \left(x^{3/2} - x^3 - 2\sqrt{x} + 2x^2\right) dx$$

$$= -\left[\frac{x^{\frac{3}{2}+1}}{5/2} - \frac{x^4}{4} - 2\frac{x^{\frac{1}{2}+1}}{3/2} + \frac{2x^3}{3}\right]_0^1$$

$$= -\left[\frac{2}{5}x^{5/2} - \frac{x^4}{4} - 2\times\frac{2}{3}x^{3/2} + \frac{2x^3}{3}\right]_0^1$$

$$= -\left[\frac{2}{5}(1)^{5/2} - \frac{(1)^4}{4} - \frac{4(1)^{3/2}}{3} + \frac{2(1)^3}{3}\right] - 0$$

$$= -\left[\frac{2}{5} - \frac{1}{4} - \frac{4}{3} + \frac{2}{3}\right] \Rightarrow -\left[\frac{24-15-80+40}{60}\right]$$

$$= -\left[\frac{-31}{60}\right] \Rightarrow \frac{31}{60} \quad \text{Ans.}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

In special case when $P = y$, $Q = -x$, so

$$\frac{\partial P}{\partial y} = 1, \quad \frac{\partial Q}{\partial x} = -1$$

Green Theorem states that

$$+\iint_R \left(\frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x}\right) dx\,dy = \oint_C (P\,dx + Q\,dy)$$

$$\iint_R \left(1 - (-1)\right) dx\,dy = -\oint_C (y\,dx - x\,dy)$$

$$2\iint_R dx\,dy = \oint_C (x\,dy - y\,dx)$$

$$\left(\text{Area of closed region}\right) \iint_R dx\,dy = \frac{1}{2}\oint_C (x\,dy - y\,dx)$$

Example:- Determine the area of figure enclosed by $y = 3x^2$ & $y = 6x$. ~~Points of intersection~~

~~$3x^2 =$~~

$3x^2 = 6x$

$3x^2 - 6x = 0$

$3x(x-2) = 0$

$3x = 0$ | $x - 2 = 0$

$x = 0$ | $x = 2$

NOTES BY KINZA BILAL

Scanned with CamScanner

For C_1 : $y = 3x^2$, $dy = 6x\,dx$

$$I_1 = \frac{1}{2}\int_0^2 (x\,dy - y\,dx)$$

$$= \frac{1}{2}\int_0^2 (x \cdot 6x\,dx - (3x^2)\,dx)$$

$$= \frac{1}{2}\int_0^2 (6x^2\,dx - 3x^2\,dx) \Rightarrow \frac{1}{2}\int_0^2 (6x^2 - 3x^2)\,dx$$

$$= \frac{1}{2}\int_0^2 3x^2\,dx \Rightarrow \frac{3}{2}\int_0^2 x^2\,dx \Rightarrow \frac{3}{2}\left[\frac{x^3}{3}\right]_0^2$$

$$= \frac{1}{2}\left[x^3\right]_0^2 \Rightarrow \frac{1}{2}(2^3 - 0^3) \Rightarrow \frac{1}{2}(8) = 4$$

For C_2 : $y = 6x$, $dy = 6\,dx$

$$I_2 = \frac{1}{2}\int_0^2 (x\,dy - y\,dx) \Rightarrow \frac{1}{2}\int_0^2 (x \cdot 6\,dx - 6x\,dx)$$

$$= \frac{1}{2}\int_0^2 (6x\,dx - 6x\,dx) = \frac{1}{2}\int_0^2 (6x - 6x)\,dx$$

$$= 0$$

Now

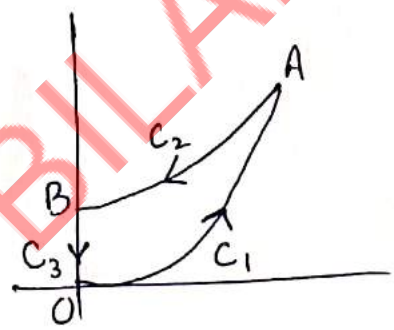
Area $= I = I_1 + I_2 \Rightarrow 4 + 0 \Rightarrow 4$ Ans.

NOTES BY KINZA BILAL

Scanned with CamScanner

Example:- Determine the area bounded by the curves $y = 2x^3$, $y = x^3 + 1$ & the axis $x = 0$ for $x \geq 0$

$y = 2x^3$, $y = x^3 + 1$, $x = 0$



put $2x^3 = x^3 + 1$

$2x^3 - x^3 = 1$

$x^3 = 1$

$\boxed{x = 1}$

I_1 • for C_1: $y = 2x^3$, $dy = 6x^2 dx$

$$I_1 = \frac{1}{2}\int_0^1 (x\,dy - y\,dx)$$

$$= \frac{1}{2}\int_0^1 \left[x(6x^2 dx) - (2x^3)\,dx\right]$$

$$= \frac{1}{2}\int_0^1 (6x^3\,dx - 2x^3 dx) \Rightarrow \frac{1}{2}\int_0^1 4x^3\,dx$$

$$= \frac{1}{2}\left[\frac{4x^4}{4}\right]_0^1 \Rightarrow \frac{1}{2}\left[x^4\right]_0^1 \Rightarrow \frac{1}{2}(1)^4 - 0^4) \Rightarrow \frac{1}{2}$$

for C_2: $y = x^3 + 1$, $dy = 3x^2 dx$

$$I_2 = \frac{1}{2}\int_1^0 (x\,dy - y\,dx) \Rightarrow \frac{1}{2}\int_1^0 \left(x(3x^2 dx) - (x^3+1)dx\right]$$

$$= \frac{1}{2}\int_1^0 (3x^3 - x^3 - 1)\,dx \Rightarrow \frac{1}{2}\int_1^0 (2x^3 - 1)\,dx$$

$$= \frac{1}{2}\left[\frac{2x^4}{4} - x\right]_1^0 \Rightarrow \frac{1}{2}\left[0 - \left(\frac{(1)^4}{2} - 1\right)\right] \Rightarrow \frac{1}{2}\left[-\left(\frac{1-2}{2}\right)\right]$$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$= \frac{1}{2}\left[-\left(-\frac{1}{2}\right)\right] \Rightarrow \frac{1}{2}\left(\frac{1}{2}\right) \Rightarrow \frac{1}{4}$$

for C_3 : $x = 0$, $dx = 0$

$$I_3 = \frac{1}{2}\oint_C (x\,dy - y\,dx)$$

$$I_3 = 0$$

Now Area $= I = I_1 + I_2 + I_3$

$$= \frac{1}{2} + \frac{1}{4} + 0 \Rightarrow \frac{2+1}{4} = \frac{3}{4}$$ sq. units

Revision Summary:

Properties of line integral:-

1. Sign of integral is reversed when direction of integral is reversed.
2. Path of integration parallel to y-axis, $dx = 0$

   $\therefore I_c = \int_c Q\,dy$
3. Path of integration parallel to x-axis, $dy = 0$

   $\therefore I_c = \int_c P\,dx$
4. Path of integration must be continuous & single valued.
5. Dependence of line integral on path of integration.

NOTES BY KINZA BILAL

Scanned with CamScanner

6. The value of line integral depends on particular path of integration.

7. Exact differential

- If $Pdx + Qdy$ is an exact differential

(a) $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$

(b) $I = \int_C (Pdx + Qdy)$ is independent of path.

(c) $I = \oint_C (Pdx + Qdy)$ is zero.

8. Exact differential in three variables.

- If $Pdx + Qdy + Rdw$ is an exact differential

(a) $\frac{\partial P}{\partial y} = \frac{\partial Q}{\partial x}$, $\frac{\partial P}{\partial w} = \frac{\partial R}{\partial x}$, $\frac{\partial R}{\partial y} = \frac{\partial Q}{\partial w}$

(b) $\int_C (P(dx + Qdy + Rdw)$ is independent of path.

(c) $\oint_C (Pdx + Qdy + Rdw)$ is zero.

9. Green's Theorem.

$$\oint_C (Pdx + Qdy) = -\iint_R \left\{ \frac{\partial P}{\partial y} - \frac{\partial Q}{\partial x} \right\} dx\,dy$$ & for simple closed curve $\oint_C (x\,dy - y\,dx) = 2\iint_R dx\,dy = 2A$

where A is area of closed figure.

NOTES BY KINZA BILAL

Scanned with CamScanner

Gradient of Scalar function:-

Del operator is given by $\nabla = \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k\right)$

$$\nabla\phi = \text{grad}\,\phi = \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k\right)\phi$$

$$\nabla\phi = \text{grad}\,\phi = \frac{\partial\phi}{\partial x} i + \frac{\partial\phi}{\partial y} j + \frac{\partial\phi}{\partial z} k$$

Div (Divergence of vector function).

If $A = a_1 i + a_2 j + a_3 k$, Then

$$\text{div } A = \nabla \cdot A = \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k\right) \cdot \left(a_1 i + a_2 j + a_3 k\right)$$

$$= \frac{\partial a_1}{\partial x} + \frac{\partial a_2}{\partial y} + \frac{\partial a_3}{\partial z}$$

Note That

(a) grad operator ∇ acts on scalar & gives vector.

(b) div operator $\nabla\cdot$ acts on vector & gives scalar.

Example:- If $A = x^2 y i - xyz j + yz^2 k$. find div A.

$$\text{div } A = \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k\right) \cdot \left(x^2 y i - xyz j + yz^2 k\right)$$

$$= \frac{\partial}{\partial x}(x^2 y) + \frac{\partial}{\partial y}(-xyz) + \frac{\partial}{\partial z}(yz^2)$$

$$\nabla \cdot A = 2xy - xz + 2yz$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Example:- If $A = 2x^2y\,i - 2(xy^2 + y^3z)j + 3y^2z^2k$ determine $\nabla.A$ (div A).

$$\nabla.A = \frac{\partial}{\partial x}(2x^2y) + \frac{\partial}{\partial y}(-2(xy^2 + y^3z)) + \frac{\partial}{\partial z}(3y^2z^2)$$

$$= 4xy - 2(2xy + 3y^2z) + 6y^2z$$

$$= 4xy - 4xy - 6y^2z + 6y^2z \Rightarrow 0$$

vector for which $\nabla.A = 0$ called a Solenoid vector.

Curl (curl of a vector function) :-

Curl operator denoted by $\nabla\times$ acts on vector & also gives a vector.

If $A = a_1\hat{i} + a_2 j + a_3 k$ then curl $A = \nabla \times A$

$$= \left(\frac{\partial}{\partial x}i + \frac{\partial}{\partial y}j + \frac{\partial}{\partial z}k\right) \times (a_1\hat{i} + a_2 j + a_3 k)$$

$$= \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a_1 & a_2 & a_3 \end{vmatrix}$$

$$\nabla \times A = i\left(\frac{\partial a_3}{\partial y} - \frac{\partial a_2}{\partial z}\right) - j\left(\frac{\partial a^3}{\partial x} - \frac{\partial a_1}{\partial z}\right) + k\left(\frac{\partial a_2}{\partial x} - \frac{\partial a_1}{\partial y}\right)$$

curl A is a vector function.

NOTES BY KINZA BILAL

Scanned with CamScanner

Example:- If $A = (y^4 - x^2z^2)i + (x^2+y^2)j - x^2yz\,k$, determine curl A at (1, 3, −2)

$$\text{Curl } A = \nabla \times A = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ y^4 - x^2z^2 & x^2+y^2 & -x^2yz \end{vmatrix}$$

$$= i\left(\frac{\partial}{\partial y}(-x^2yz) - \frac{\partial}{\partial z}(x^2+y^2)\right) - j\left(\frac{\partial}{\partial x}(-x^2yz) - \frac{\partial}{\partial z}(y^4 - x^2z^2)\right)$$

$$+ k\left(\frac{\partial}{\partial x}(x^2+y^2) - \frac{\partial}{\partial y}(y^4 - x^2z^2)\right)$$

$$= i(-x^2z - 0) - j(-2xyz + 2x^2z) + k(2x - 4y^3)$$

= put (1, 3, −2)

$$= i(-(1)^2(-2)) - j[-2(1)(3)(-2) + 2(1)^2(-2)] + k(2(1) - 4(3)^3)$$

$$= i(2) - j(12-48) + k(2-108)$$

$$= 2i - 8j - 106k \quad \text{Ans.}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Example: Determine curl F at point (2,0,3) given that $F = ze^{2xy}\,i + 2xz\cos y\,j + (x+2y)k$.

$$\nabla \times F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ ze^{2xy} & 2xz\cos y & x+2y \end{vmatrix}$$

$$= i\left[2 - 2x\cos y\right] - j\left[1 - e^{2xy}\right] + k\left[2z\cos y - ze^{2xy}\cdot 2x\right]$$

Put (2, 0, 3)

$$= i\left[2 - 2(2)\cos 0^\circ\right] - j\left[1 - e^{2(2)(0)}\right] + k\left[2(3)\cos 0^\circ - (3)e^{2(2)(0)}\cdot 2(2)\right]$$

$$= i(-2) - j(1-1) + k(6-12)$$

$$= -2i - 6k \Rightarrow -2(i + 3k).$$

Summary of grad, div, curl

(a) grad ∇ acts on scalar to give vector.

(b) Div $\nabla\cdot$ acts on vector to give scalar.

(c) Curl $\nabla\times$ acts on vector to give vector.

(d) with a scalar function. $\phi(x,y,z)$

$$\text{grad}\,\phi = \nabla\phi = \frac{\partial\phi}{\partial x}i + \frac{\partial\phi}{\partial y}j + \frac{\partial\phi}{\partial z}k$$

NOTES BY KINZA BILAL

Scanned with CamScanner

(e) With a vector function $A = a_1 i + a_2 j + a_3 k$

(i) $\text{div } A = \nabla . A = \frac{\partial a_1}{\partial x} + \frac{\partial a_2}{\partial y} + \frac{\partial a_3}{\partial z}$

(ii) $\text{curl } A = \nabla \times A = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a_1 & a_2 & a_3 \end{vmatrix}$

Multiple Operations:-

We can combine the operators grad, div & curl in multiple operations.

Example:- $A = x^2 y i + y z^3 j + z x^3 k$

div A = ?, grad (div A) = ?

$$\text{div } A = \nabla . A = \frac{\partial}{\partial x}(x^2 y) + \frac{\partial}{\partial y}(y z^3) + \frac{\partial}{\partial z}(z x^3)$$

$$= 2xy + z^3 + x^3$$

Now

$$\text{grad (div } A) = \nabla (\nabla . A) = \frac{\partial}{\partial x}(\text{div } A) + \frac{\partial}{\partial y}(\text{div } A) + \frac{\partial (\text{div } A)}{\partial z}$$

$$= \frac{\partial}{\partial x}(2xy + z^3 + x^3) i + \frac{\partial}{\partial y}(2xy + z^3 + x^3) j + \frac{\partial}{\partial z}(2xy + z^3 + x^3) k$$

$$= (2y + 3x^2) i + 2x j + 3z^2 k \quad \underline{\text{Ans.}}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Example:- If $\phi = xyz - 2y^2z + x^2z^2$.

find div grad ϕ at $(2,4,1)$

$$\text{grad}\,\phi = \nabla\phi = \frac{\partial}{\partial x}\phi i + \frac{\partial}{\partial y}\phi j + \frac{\partial}{\partial z}\phi k$$

$$= (yz - 2xz^2)i + (xz - 4yz)j + (xy - 2y^2 + 2x^2z)k$$

Now

$$\text{div grad}\,\phi = \nabla\cdot(\nabla\phi) = \frac{\partial}{\partial x}a_1 + \frac{\partial}{\partial y}a_2 + \frac{\partial}{\partial z}a_3$$

$$= 2z^2 - 4z + 2x^2$$

put $(2,4,1)$

$$= 2(1)^2 - 4(1) + 2(2)^2$$

$$= 2 - 4 + 8 \Rightarrow 6 \quad \text{Ans.}$$

$$\therefore \text{div grad}\,\phi = \frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2}$$

Example:- If $F = x^2yz\,i + xyz^2\,j + y^2z\,k$

determine curl F at the point $(2,1,1)$. Determine an expression for curl F in the usual way, which will be a vector, & then curl of the result. Finally substitute values. (i.e $\nabla \times (\nabla \times F)$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$\text{curl } F = \nabla \times F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ x^2yz & xyz^2 & y^2z \end{vmatrix}$$

$$= i(2yz - 2xyz) - j(0 - x^2y) + k(yz^2 - x^2z)$$

Now

$$\nabla \times (\nabla \times F) = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2yz - 2xyz & +x^2y & yz^2 - x^2z \end{vmatrix}$$

$$= i[(z^2 - 0) - 0] - j[(0 - 2xz) - (2y - 2xy)] + k[2xy - (2y - 2xz)$$

$$= z^2 i - j(-2xz - 2y + 2xy) + k(2xy - 2y + 2xz)$$

Put (2, 1, 1)

$$= (1)^2 i - j[-2(2)(1) - 2(1) + 2(2)(1)] + k[2(2)(1) - 2(1) + 2(2)(1)$$

$$= i - j[-4 - 2 + 4] + k[4 - 2 + 4]$$

$$= i - j(-2) + k(6)$$

$$= i + 2j + 6k \quad \text{Ans.}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Two intersecting general results-

(a) Curl grad ϕ where ϕ is scalar.

$$\text{Curl grad } \phi = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ \frac{\partial \phi}{\partial x} & \frac{\partial \phi}{\partial y} & \frac{\partial \phi}{\partial z} \end{vmatrix}$$

$$= i\left[\frac{\partial^2\phi}{\partial y \partial z} - \frac{\partial^2\phi}{\partial z \partial y}\right] - j\left[\frac{\partial^2\phi}{\partial x \partial z} - \frac{\partial^2\phi}{\partial z \partial x}\right] + k\left[\frac{\partial^2\phi}{\partial x \partial y} - \frac{\partial^2\phi}{\partial y \partial x}\right]$$

$$= 0$$

So curl grad ϕ

$$\nabla \times (\nabla \phi) = 0$$

(b) div Curl A where A is vector

$$\text{Curl } A = \nabla \times A = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ a_1 & a_2 & a_3 \end{vmatrix}$$

$$= i\left(\frac{\partial a_3}{\partial y} - \frac{\partial a_2}{\partial z}\right) - j\left(\frac{\partial a_3}{\partial x} - \frac{\partial a_1}{\partial z}\right) + k\left(\frac{\partial a_2}{\partial x} - \frac{\partial a_1}{\partial y}\right)$$

$$\text{div Curl } A = \nabla \cdot (\nabla \times A)$$

$$= \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k\right) \cdot (\nabla \times A)$$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$= \frac{\partial^2 a_3}{\partial x \partial y} - \frac{\partial^2 a_2}{\partial z \partial x} - \frac{\partial^2 a_3}{\partial x \partial y} + \frac{\partial^2 a_1}{\partial y \partial z} - \frac{\partial^2 a_2}{\partial z \partial x} - \frac{\partial^2 a_1}{\partial y \partial z}$$

$$= 0$$

So $\boxed{\text{div curl } A = \nabla \cdot (\nabla \times A) = 0}$

(c) div grad ϕ where ϕ is scalar.

$$\nabla\phi = \text{grad } \phi = \frac{\partial \phi}{\partial x} i + \frac{\partial \phi}{\partial y} j + \frac{\partial \phi}{\partial z} k$$

Now

$$\text{div grad } \phi = \nabla \cdot (\nabla \phi)$$

$$= \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k\right) \cdot \left(\frac{\partial \phi}{\partial x} i + \frac{\partial \phi}{\partial y} j + \frac{\partial \phi}{\partial z} k\right)$$

$$= \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2}$$

$$= \nabla^2 \phi$$

So

$$\boxed{\text{div grad } \phi = \nabla \cdot (\nabla \phi) = \frac{\partial^2 \phi}{\partial x^2} + \frac{\partial^2 \phi}{\partial y^2} + \frac{\partial^2 \phi}{\partial z^2} = \nabla^2 \phi}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Lecture 35

Definite Integrals

for $\sin^n x$ & $\cos^n x$, $\left(0 \le x \le \frac{\pi}{2}\right)$

$$\sin\frac{x}{2} = \sqrt{\frac{1-\cos x}{2}}$$

$$\sin^2\frac{x}{2} = \frac{1-\cos x}{2}$$

$$\sin^2 x = 1-\cos 2x$$

$$\int_0^{\pi/2} \sin^2 x\,dx = \int_0^{\pi/2} \left(\frac{1-\cos 2x}{2}\right) dx$$

$$= \frac{1}{2}\int_0^{\frac{\pi}{2}} (1-\cos 2x)\,dx$$

$$= \frac{1}{2}\int_0^{\frac{\pi}{2}} (1-\cos 2x)\,dx = \frac{1}{2}\left[x - \frac{\sin 2x}{2}\right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{2}\left[\left\{\frac{\pi}{2} - \frac{\sin 2\left(\frac{\pi}{2}\right)}{2}\right\} - \left\{0 - \frac{\sin 2(0)}{2}\right\}\right]$$

$$= \frac{1}{2}\left[\frac{\pi}{2} - 0 - 0 + 0\right] \Rightarrow \frac{1}{2}\left(\frac{\pi}{2}\right)$$

$$= \frac{1}{2}\cdot\frac{\pi}{2} = \frac{\pi}{4} \quad \text{Ans.}$$

$$\int_0^{\frac{\pi}{2}} \cos^2 x\,dx = \int_0^{\frac{\pi}{2}} \left(\frac{1+\cos 2x}{2}\right) dx \Rightarrow \frac{1}{2}\int_0^{\frac{\pi}{2}} (1+\cos 2x)$$

$$= \frac{1}{2}\left[x + \frac{\sin 2x}{2}\right]_0^{\frac{\pi}{2}} \Rightarrow \frac{1}{2}\left[\left\{\frac{\pi}{2} + \frac{\sin 2\left(\frac{\pi}{2}\right)}{2}\right\} - \{0\}\right]$$

NOTES BY KINZA BILAL

Scanned with CamScanner

$= \frac{1}{2}\left(\frac{\pi}{2}\right) \Rightarrow \frac{1}{2}\cdot\frac{\pi}{2} \Rightarrow \frac{\pi}{4}$ Ans.

$$\int_0^{\pi/2} \sin^3 x\,dx = \int_0^{\pi/2} \sin^2 x \sin x\,dx$$

$$= \int_0^{\pi/2} (1-\cos^2 x)\sin x\,dx \Rightarrow \int_0^{\pi/2} (\sin x - \cos^2 x \sin x)\,dx$$

$$= \int_0^{\pi/2} \sin x\,dx + \int_0^{\pi/2} \cos^2 x\,(-\sin x)\,dx$$

$$= \left[-\cos x\right]_0^{\frac{\pi}{2}} + \left[\frac{\cos^3 x}{3}\right]_0^{\frac{\pi}{2}}$$

$$= -\left[\cos\frac{\pi}{2} - \cos 0^\circ\right] + \frac{1}{3}\left[\cos^3\frac{\pi}{2} - \cos^3 0\right]$$

$$= -\left[0 - 1\right] + \frac{1}{3}\left[(0)^3 - (1)^3\right]$$

$$= 1 + \frac{1}{3}(-1) \Rightarrow 1 - \frac{1}{3} = \frac{3-1}{3} = \frac{2}{3}$$ Ans.

$$\int_0^{\pi/2} \cos^3 x\,dx = \int_0^{\pi/2} \cos^2 x \cos x\,dx = \int_0^{\pi/2} (1-\sin^2 x)\cos x\,dx$$

$$= \int_0^{\pi/2} (\cos x - \sin^2 x \cos x)\,dx \Rightarrow \left[\sin x\right]_0^{\pi/2} - \left[\frac{\sin^3 x}{3}\right]_0^{\frac{\pi}{2}}$$

$$= \left(\sin\frac{\pi}{2} - \sin 0\right) - \frac{1}{3}\left(\sin^3\frac{\pi}{2} - \sin^3 0\right)$$

$$= (1-0) - \frac{1}{3}\left((1)^3 - (0)^3\right) \Rightarrow 1 - \frac{1}{3} \Rightarrow \frac{2}{3}$$ Ans.

NOTES BY KINZA BILAL

Scanned with CamScanner

$$\int_0^{\pi/2} \sin^4 x\,dx = \int_0^{\pi/2} (\sin^2 x)^2\,dx \Rightarrow \int_0^{\pi/2} \left[\frac{1-\cos 2x}{2}\right]^2$$

$$= \int_0^{\frac{\pi}{2}} \left[\frac{1 - 2\cos 2x + \cos^2 2x}{4}\right] dx$$

$$= \frac{1}{4}\int_0^{\frac{\pi}{2}} \left(1 - 2\cos 2x + \frac{1+\cos 4x}{2}\right) dx$$

$$= \frac{1}{4}\int_0^{\frac{\pi}{2}} \left(1 - 2\cos 2x + \frac{1}{2} + \frac{\cos 4x}{2}\right) dx$$

$$= \frac{1}{4}\int_0^{\pi/2} \left(\frac{3}{2} - 2\cos 2x + \frac{\cos 4x}{2}\right) dx$$

$$= \frac{1}{4}\left[\frac{3}{2}x - 2\,\frac{\sin 2x}{2} + \frac{\sin 4x}{2\times 4}\right]_0^{\frac{\pi}{2}}$$

$$= \frac{1}{4}\left[\frac{3}{2}\left(\frac{\pi}{2}\right) - \sin 2\left(\frac{\pi}{2}\right) + \frac{\sin 4\left(\frac{\pi}{2}\right)}{8}\right] - 0$$

$$= \frac{1}{4}\left(\frac{3}{2}\cdot\frac{\pi}{2} - 0 + 0\right) \Rightarrow \frac{1}{4}\cdot\frac{3}{2}\cdot\frac{\pi}{2}$$

$$= \frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2}$$

$$\cos\frac{x}{2} = \sqrt{\frac{1+\cos x}{2}}$$

$$\cos^2\frac{x}{2} = \frac{1+\cos x}{2}$$

$$\cos^2 x = \frac{1+\cos 2x}{2}$$

$$\cos^2 2x = \frac{1+\cos 4x}{2}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$\int_0^{\pi/2} \cos^4 x\,dx = \int_0^{\pi/2} (\cos^2 x)^2\,dx = \int_0^{\pi/2} \left[\frac{1+\cos 2x}{2}\right]^2 dx$$

$$= \int_0^{\frac{\pi}{2}} \left(\frac{1+2\cos 2x+\cos^2 2x}{4}\right) dx$$

$$= \frac{1}{4}\int_0^{\frac{\pi}{2}} \left(1+2\cos 2x+\frac{1+\cos 4x}{2}\right) dx$$

$$= \frac{1}{4}\int_0^{\pi/2} \left(1+2\cos 2x+\frac{1}{2}+\frac{\cos 4x}{2}\right) dx$$

$$= \frac{1}{4}\int_0^{\pi/2} \left(\frac{3}{2}+2\cos 2x+\frac{\cos 4x}{2}\right) dx$$

$$= \frac{1}{4}\left[\frac{3}{2}x+\frac{2\sin 2x}{2}+\frac{\sin 4x}{8}\right]_0^{\pi/2}$$

$$= \frac{1}{4}\left[\frac{3}{2}\left(\frac{\pi}{2}\right)+\frac{2\sin 2\left(\frac{\pi}{2}\right)}{2}+\frac{\sin 4\left(\frac{\pi}{2}\right)}{8}\right]-0$$

$$= \frac{1}{4}\left[\frac{3}{2}\cdot\frac{\pi}{2}+0+0\right]$$

$$= \frac{1}{4}\cdot\frac{3}{2}\cdot\frac{\pi}{2}$$

$$= \frac{3}{4}\cdot\frac{1}{2}\cdot\frac{\pi}{2} \quad \text{Ans.}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$\Rightarrow \int_0^{\pi/2} \sin^5 x\,dx = \frac{4}{5}\cdot\frac{2}{3}, \quad \int_0^{\pi/2} \cos^5 x\,dx = \frac{4}{5}\cdot\frac{2}{3}$$

$$\Rightarrow \int_0^{\pi/2} \sin^6 x\,dx = \frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}, \quad \int_0^{\pi/2} \cos^6 dx = \frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}$$

$$\Rightarrow \int_0^{\pi/2} \sin^7 x\,dx = \frac{6}{7}\frac{4}{5}\frac{2}{3}, \quad \int_0^{\pi/2} \cos^7 x\,dx = \frac{6}{7}\frac{4}{5}\frac{2}{3}$$

$$\Rightarrow \int_0^{\pi/2} \sin^8 x\,dx = \frac{7}{8}\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}, \quad \int_0^{\pi/2} \cos^8 x\,dx = \frac{7}{8}\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}$$

$$\Rightarrow \int_0^{\pi/2} \sin^9 x\,dx = \frac{8}{9}\frac{6}{7}\frac{4}{5}\frac{2}{3}, \quad \int_0^{\pi/2} \cos^9 x\,dx = \frac{8}{9}\frac{6}{7}\frac{4}{5}\frac{2}{3}$$

$$\Rightarrow \int_0^{\pi/2} \sin^{10} x\,dx = \frac{9}{10}\frac{7}{8}\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}, \quad \int_0^{\pi/2} \cos^{10} x\,dx = \frac{9}{10}\frac{7}{8}\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}$$

Wallis sine formula:-

When n is even:

$$\int_0^{\pi/2} \sin^n x\,dx = \frac{n-1}{n}\cdot\frac{n-3}{n-2}\cdot\frac{n-5}{n-4}\cdot\frac{n-7}{n-6}\cdots\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}$$

When n is odd:

$$\int_0^{\pi/2} \cos^n x\,dx = \frac{n-1}{n}\cdot\frac{n-3}{n-2}\cdot\frac{n-5}{n-4}\cdot\frac{n-7}{n-6}\cdots\frac{6}{7}\frac{4}{5}\frac{2}{3}$$

$$\Rightarrow \int_0^{\pi/2} \sin^{11} x\,dx = \frac{10}{11}\cdot\frac{8}{9}\cdot\frac{6}{7}\cdot\frac{4}{5}\cdot\frac{2}{3}, \quad \int_0^{\pi/2} \cos^{11} x\,dx = \frac{10}{11}\cdot\frac{8}{9}\cdot\frac{6}{7}\cdot\frac{4}{5}\cdot\frac{2}{3}$$

$$\Rightarrow \int_0^{\pi/2} \sin^{12} x\,dx = \frac{11}{12}\cdot\frac{9}{10}\cdot\frac{7}{8}\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}, \quad \int_0^{\pi/2} \cos^{12} x\,dx = \frac{11}{12}\frac{9}{10}\frac{7}{8}\frac{5}{6}\frac{3}{4}\frac{1}{2}\frac{\pi}{2}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Integration by parts:-

$$\int UV\,dx = U\int V\,dx - \int\left(\int V\,dx \cdot \frac{dU}{dx}\right)dx$$

Example:- Evaluate $\int x \ln x \cdot dx$

$$\int \ln x \cdot x\,dx = \ln x \int x\,dx - \int\left(\int x\,dx \cdot \frac{d(\ln x)}{dx}\right)dx$$

$$= \ln x \cdot \frac{x^2}{2} - \int\left(\frac{x^2}{2}\cdot\frac{1}{x}\right)dx$$

$$= \ln x \cdot \frac{x^2}{2} - \int \frac{x}{2}\,dx \Rightarrow \ln x \cdot \frac{x^2}{2} - \frac{1}{2}\int x\,dx$$

$$= \ln x \cdot \frac{x^2}{2} - \frac{1}{2}\cdot\frac{x^2}{2} \Rightarrow \frac{x^2}{2}\left[\ln x - \frac{1}{2}\right] \text{ Ans}$$

Example:- Evaluate $\int x \sin x\,dx$

$$\int x \sin x\,dx = x\int \sin x\,dx - \int\left(\int \sin x\,dx \cdot \frac{d(x)}{dx}\right)dx$$

$$= x(-\cos x) - \int(-\cos x \cdot 1)\,dx$$

$$= -x\cos x + \int \cos x\,dx$$

$$= -x\cos x + \sin x \quad \text{Ans.}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Line integrals

length of curve $AB = \int_c dr$ when $dr \to 0$

Lecture 36 Scalar Field

If a scalar field $V(r)$ exist for all points on the curve, Then line integral $= \int_c V(r)\,dr$.

To evaluate line integral, integrand is expressed in terms of x, y, z with $dr = dx\,i + dy\,j + dz\,k$

Example:- If $V = xy^2z$, evaluate $\int_c V(r)\,dr$ along the curve c having parametric equations $x = 3u$, $y = 2u^2$, $z = u^3$ between $A(0,0,0)$ & $B = (3,2,1)$.

Solution:- $V = xy^2z = (3u)(2u^2)^2(u^3)$

$= (3u)(4u^4)(u^3)$

$= 12u^8$

$x = 3u$, $dx = 3$

$y = 2u^2$, $dy = 4u$

$z = u^3$, $dz = 3u^2$

$dr = dx\,i + dy\,j + dz\,k$

$= 3\,du\,i + 4u\,du\,j + 3u^2 k\,du$

$= (3i + 4u\,j + 3u^2 k)\,du$

put $A(0,0,0)$ in $0 = 3u$, $0 = 2u^2$, $0 = u^3$

$0 = u$, $0 = u$, $0 = u$

put $B(3,2,1)$, $3 = 3u$, $2 = 2u^2$, $1 = u^3$

$1 = u$, $1 = u$, $1 = u$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$\int_c V(r)\,dr = \int_0^1 12u^8(3i+4uj+3u^2k)\,du$$

$$= \int_0^1 (36u^8 i + 48u^9 j + 36u^{10} k)\,du$$

$$= \left[36\frac{u^9}{9}i + 48\frac{u^{10}}{10}j + 36\frac{u^{11}}{11}k\right]_0^1$$

$$= \left(4(1)^9 i + \frac{24}{5}(1)^{10} j + \frac{36}{11}(1)^{11} k\right) - 0$$

$$= 4i + \frac{24}{5}j + \frac{36}{11}k \quad \text{Ans.}$$

Example:- If $V = xy + y^2z$. Evaluate $\int_c V(r)\,dr$ along curve c defined by $x=t^2$, $y=2t$, $z=t+5$ between $(0,0,5)$ & $(4,4,7)$.

$$V = xy + y^2z = (t^2)(2t) + (2t)^2(t+5)$$

$$= 2t^3 + 4t^2(t+5)$$

$$= 2t^3 + 4t^3 + 20t^2$$

$$= 6t^3 + 20t^2$$

Now

$x = t^2$, $dx = 2t\,dt$

$y = 2t$, $dy = 2\,dt$

$z = t+5$, $dz = 1\,dt$

$$dr = dx\,i + dy\,j + dz\,k = 2t\,dt\,i + 2dt\,j + dt\,k$$

$$= (2t\,i + 2j + k)\,dt$$

NOTES BY KINZA BILAL

Scanned with CamScanner

put (0,0,5) in

$x = t^2$	$y = 2t$	$z = t+5$
$0 = t^2$	$0 = 2t$	$5 = t+5$
$0 = t$	$0 = t$	$0 = t$

put (4,4,7) in

$4 = t^2$	$4 = 2t$	$7 = t+5$
$2 = t$	$\frac{4}{2} = t$	$7-5 = t$
	$2 = t$	$2 = t$

So

$$\int_C V(r)\,dr = \int_0^2 (6t^3 + 20t^2)(2t\,i + 2j + k)\,dt$$

$$= \int_0^2 (12t^4 i + 6t^3 j + 40t^3 i + 12t^3 j + 40t^2 j + 6t^3 k + 20t^2 k)\,dt$$

$$= 2\int_0^2 (6t^4 + 20t^3)i + (6t^3 + 20t^2)j + (3t^3 + 10t^2)k$$

$$= 2\left[\left(\frac{6t^5}{5} + \frac{20t^4}{4}\right)i + \left(\frac{6t^4}{4} + \frac{20t^3}{3}\right)j + \left(\frac{3t^4}{4} + \frac{10t^3}{3}\right)k\right]_0^2$$

$$= 2\left[\left(\frac{6(2)^5}{5} + \frac{20(2)^4}{4}\right)i + \left(\frac{6(2)^4}{4} + \frac{20(2)^3}{3}\right)j + \left(\frac{3(2)^4}{4} + \frac{10(2)^3}{3}\right)\right]$$

$$= 2\left[\left(\frac{6(32)}{5} + \frac{20(16)}{4}\right)i + \left(\frac{6(16)}{4} + \frac{20(8)}{3}\right)j + \left(\frac{3(16)}{4} + \frac{10(8)}{3}\right)\right]$$

$$= 2\left[\left(\frac{192}{5} + 80\right)i + \left(24 + \frac{160}{3}\right)j + \left(12 + \frac{80}{3}\right)k\right]$$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$= 2\left[\left(\frac{192+400}{5}\right)i + \left(\frac{72+160}{3}\right)j + \left(\frac{36+80}{3}\right)k \right]$$

$$= 2\left[\frac{592}{5}i + \frac{232}{3}j + \frac{116}{3}k \right]$$

$$= 2\times 4\left[\frac{148}{5}i + \frac{58}{3}j + \frac{29}{3}k \right]$$

$$= 8\left[\frac{444i + 290j + 145k}{15} \right]$$

$$= \frac{8}{15}\left[444i + 290j + 145k \right]$$ Ans.

Vector Field:-

If a vector field $F(r)$ exists for all points of the curve c, Then for each element of arc

The line integral $= \int_c F.dr$

here $F = F_1 i + F_2 j + F_3 k$ & $dr = dx\,i + dy\,j + dz\,k$

Then $\int_c F.dr = \int_c (F_1 i + F_2 j + F_3 k).(dx\,i + dy\,j + dz\,k)$

$$\int_c F.dr = \int_c F_1 dx + F_2 dy + F_3 dz$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Example:- If $F(r) = x^2yi + xzj + 2yzk$,

Evaluate $\int_c F.dr$ b/w $A(0,0,0)$ & $B(4,2,1)$ along curve c having parametric equations

$x = 4t, \quad y = 2t^2, \quad z = t^3$

$dx = 4dt, \quad dy = 4t\,dt, \quad dz = 3t^2dt$

$$F(r) = x^2yi + xzj + 2yzk$$

$$= (4t)^2(2t^2)\,i + (4t)(t^3)j + 2(2t^2)(t^3)k$$

$$= 32t^4 i + 4t^4 j + 4t^5 k$$

& $dr = (4i + 4tj + 3t^2k)\,dt$

Now

$$\int_c F.dr = \int_c (32t^4 i + 4t^4 j + 4t^5 k).(4i + 4tj + 3t^2 k)\,dt$$

$$= \int_c (128t^4 + 16t^5 + 12t^7)\,dt$$

Limits:- put $(0,0,0)$ in

$x = 4t$	$y = 2t^2$	$z = t^3$
$0 = 4t$	$0 = 2t^2$	$0 = t^3$
$0 = t$	$0 = t$	$0 = t$

put $(4,2,1)$

$4 = 4t$	$2 = 2t^2$	$1 = t^3$
$1 = t$	$\frac{2}{2} = t^2$	$1 = t$
	$1 = t$	

NOTES BY KINZA BILAL

Scanned with CamScanner

$$= \int_c (128t^4 + 16t^5 + 12t^7)\,dt$$

$$= \int_0^1 \left(128\frac{t^4}{5} + 16t^5 + 12t^7\right) dt$$

$$= \left[128\frac{t^5}{5} + 16\frac{t^6}{6} + 12\frac{t^8}{8}\right]_0^1$$

$$= \left(\frac{128(1)^5}{5} + \frac{16(1)^6}{6} + \frac{12(1)^8}{8}\right) - 0$$

$$= \frac{128}{5} + \frac{16}{6} + \frac{12}{8} \Rightarrow 25.6 + 2.67 + 1.5$$

$$\int_c F.dr = 29.76 \quad \text{Ans.}$$

Example:- If $F(r) = x^2y\,i + 2yz\,j + 3z^2x\,k$.

Evaluate $\int_c F.dr$ b/w A(0,0,0) & B(1,2,3)

(a) along straight line

c_1 from (0,0,0) to (1,0,0)

c_2 from (1,0,0) to (1,2,0)

c_3 from (1,2,0) to (1,2,3)

(b) along straight line c_4 joining (0,0,0) to (1,2,3).

NOTES BY KINZA BILAL

Scanned with CamScanner

$$\int_C F \cdot dr = \int_C (x^2y\,i + 2yz\,j + 3z^2x\,k) \cdot (dx\,i + dy\,j + dz\,k)$$

$$= \int_C (x^2y\,dx + 2yz\,dy + 3z^2x\,dz)$$

(i) C_1, $y=0$, $z=0$, $dy=0$, $dz=0$

$$\int_{C_1} F \cdot dr = 0+0+0 \Rightarrow \underline{0}$$

(ii) C_2 $x=1$, $z=0$, $dx=0$, $dz=0$

$$\int_{C_2} F \cdot dr = 0+0+0 \Rightarrow \underline{0}$$

(iii) C_3

$x=1$, $y=2$, $dx=0$, $dy=0$

$$\int_{C_3} F \cdot dr = \int_{C_3} 0+0+3z^2(1)\,dz$$

$$= \int_0^3 3z^2\,dz \Rightarrow \left[\frac{3z^3}{3}\right]_0^3$$

$$= \left[z^3\right]_0^3 \Rightarrow (3)^3 - (0)^3 = \underline{27}$$

Add all

$$\int_{(0,0,0)}^{(1,2,3)} F \cdot dr = 0+0+27 \Rightarrow 27 \quad \underline{\text{Ans.}}$$

$$\therefore \int_{C_1+C_2+C_3} F \cdot dr = 27$$

NOTES BY KINZA BILAL

Scanned with CamScanner

(b) parametric equation 'll be

$x = t,\ y = 2t,\ z = 3t$ according to $(1,2,3)$

$dx = dt,\ dy = 2dt,\ dz = 3dt$

$$F(t) = x^2 y\,i + 2yz\,j + 3z^2 x\,k$$

$$= (t)^2(2t)i + 2(2t)(3t)j + 3(3t)^2(t)k$$

$$F = 2t^3 i + 12t^2 j + 27t^3 k$$

$$dr = dx\,i + dy\,j + dz\,k$$

$$= dt\,i + 2dt\,j + 3dt\,k$$

$$dr = (i + 2j + 3k)\,dt$$

Now put $(0,0,0)$ in

$x = t$	$y = 2t$	$z = 3t$
$0 = t$	$0 = 2t$	$0 = 3t$
	$0 = t$	$0 = t$

put $(1,2,3)$ in

$1 = t$	$2 = 2t$	$3 = 3t$
	$1 = t$	$1 = t$

Now $$\int_C F \cdot dr = \int_0^1 (2t^3 i + 12t^2 j + 27t^3 k) \cdot (i + 2j + 3k)\,dt$$

$$= \int_0^1 (2t^3 + 24t^2 + 81t^3)\,dt$$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$= \int_0^1 (24t^2 + 83t^3)\, dt$$

$$= \left[\frac{24t^3}{3} + \frac{83t^4}{4}\right]_0^1$$

$$= \left(\frac{24(1)^3}{3} + \frac{83(1)^4}{4}\right) - 0$$

$$= \frac{24}{3} + \frac{83}{4} \Rightarrow$$

$$= 8 + 20.75 \Rightarrow 28.75 \quad \underline{\text{Ans.}}$$

So **(a)** $\int F \cdot dr$ via $C_1, C_2, C_3 = 27$

(b) $\int F \cdot dr$ via $C_4 = 28.4$

value of line integral depends on path b/w two end points A & B.

Example:- Evaluate $\int_V F\,dv$ where v is region bounded by planes $x=0, y=0, z=0$ & $2x+y+z=2$ & $F = 2zi + yk$.

In $2x+y+z=2$

when $z=0$ $\quad 2x+y=2$, $\quad y = 2-2x$ $\quad \boxed{y=2(1-x)}$

when $y=0$ $\quad 2x+z=2$, $\quad z = 2-2x$

when $x=0$ $\quad y+z=2$, $\quad z = 2-y$

NOTES BY KINZA BILAL

Scanned with CamScanner

as $2x+y+z=2$

$z = 2-2x-y$

$\boxed{z = 2(1-x)-y}$

$2x = 2-y-z$

$2x = x-x+2x-2+2x$

$2x = 4x-2$

$2 = 4x-2x$

$2 = 2x$

$\boxed{1 = x}$

Now

$$\int_V F\,dV = \int_0^1\int_0^{2(1-x)}\int_0^{2(1-x)-y} (2z\,i + y\,k)\,dz\,dy\,dx$$

$$= \int_0^1\int_0^{2(1-x)} \left[\frac{2z^2}{2}i + yz\,k\right]_0^{2(1-x)-y}$$

$$= \int_0^1\int_0^{2(1-x)} \left[[2(1-x)-y]^2 i + y[2(1-x)-y]k\right] - 0$$

$$= \int_0^1\int_0^{2(1-x)} \left\{\left[4(1-x)^2 + y^2 - 4(1-x)y\right]i + \left[2(1-x)y - y^2\right]k\right\}$$

$$= \int_0^1 \left\{\left[4(1-x)^2 y + \frac{y^3}{3} - 4(1-x)\frac{y^2}{2}\right]i + \left[2(1-x)\frac{y^2}{2} - \frac{y^3}{3}\right]k\right\}_0^{2(1-x)}$$

$$= \int_0^1 \left\{\left(4(1-x)^2\cdot 2(1-x) + \frac{[2(1-x)]^3}{3} - 4(1-x)\frac{[2(1-x)]^2}{2}\right)i + \left(\frac{2(1-x)[2(1-x)]^2}{2} - \frac{[2(1-x)]^3}{3}\right)k\right\}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$= \int_0^1 \left\{ \left(8(1-x)^3 + \frac{8(1-x)^3}{3} - \frac{18(1-x)^3}{2} \right) i + \left(\frac{8(1-x)^3}{2} - \frac{8(1-x)^3}{3} \right) \right.$$

$$= \int_0^1 \left\{ \left(\frac{48(1-x)^3 + 16(1-x)^3 - 48(1-x)^3}{6} \right) i + \left(\frac{24(1-x)^3 - 16(1-x)^3}{6} \right) \right.$$

$$= \int_0^1 \left[\frac{16(1-x)^3}{6} i + \frac{8}{6}(1-x)^3 k \right]$$

$$= \int_0^1 \left[\frac{8}{3}(1-x)^3 i + \frac{4}{3}(1-x)^3 k \right]$$

$$= \left[\frac{8}{3} \frac{(1-x)^4}{4} i + -\frac{4}{3} \frac{(1-x)^4}{4} k \right]_0^1$$

$$= \frac{1}{3} \left[-2(1-x)^4 i + -(1-x)^4 k \right]_0^1$$

$$= \frac{1}{3} \left[0 - \left\{ -2(1)^4 i \right\} - \left\{ (-(1)^4) k \right\} \right]$$

$$= \frac{1}{3} (2i + k) \quad \text{Ans.}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Lecture 37:- Examples

Evaluate $\int_V F\,dV$ where $F = 2i + 2zj + yk$ & V is region bounded by planes $z=0$, $z=4$ & surface $x^2+y^2=9$. we'll use polar coordinates (cylindrical) (r,θ,z). bcz limits for x & y are not available

$x = r\cos\theta$, $y = r\sin\theta$, $z = z$ (comparing with (x,y,z))

$$\int_V F\,dV = \iiint_V (2i + 2zj + yk)\,dx\,dy\,dz$$

$$= \iiint_V (2i + 2zj + r\sin\theta\, k)\, r\,dr\,d\theta\,dz$$

$$= \int_0^{2\pi}\int_0^3\int_0^4 (2ri + 2rzj + r^2\sin\theta\, k)\, dz\,dr\,d\theta$$

$$= \int_0^{2\pi}\int_0^3 \left[2rzi + \frac{2rz^2}{2}j + r^2\sin\theta\, z\, k\right]_0^4 dr\,d\theta$$

$$= \int_0^{2\pi}\int_0^3 [8ri + 16rj + 4r^2\sin\theta\, k]\,dr\,d\theta$$

$$= 4\int_0^{2\pi}\int_0^3 [2ri + 4rj + r^2\sin\theta\, k]\,dr\,d\theta$$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$= 4\int_0^{2\pi} \left[\frac{2r^2}{2}i + \frac{4r^2}{2}j + \frac{r^3}{3}\sin\theta k\right]_0^3 d\theta$$

$$= 4\int_0^{2\pi} \left[9i + 18j + 9\sin\theta k\right] d\theta$$

$$= 4\times 9\int_0^{2\pi} \left[i + 2j + \sin\theta k\right] d\theta$$

$$= 36\left[\theta i + 2\theta j + (-\cos\theta)\right]_0^{2\pi}$$

$$= 36\left\{\left[2\pi i + 2(2\pi)j - \cos 2\pi k\right] - \left[0 + 0 - \cos 0 k\right]\right\}$$

$$= 36\left\{\left[2\pi i + 4\pi j - (1)k\right] - \left[-(1)k\right]\right\}$$

$$= 36\left[2\pi i + 4\pi j - k + k\right]$$

$$= 36\times 2(\pi i + 2\pi j) \Rightarrow 72(\pi i + 2\pi j) \text{ Ans.}$$

$$= 72\pi(i + 2j)$$

Scalar Field:- A Scalar field $F = xyz$ exists over curved Surface S defined by $x^2+y^2=4$ b/w The planes $z=0$ & $z=3$ in first octant.

NOTES BY KINZA BILAL

Scanned with CamScanner

Evaluate $\int_S F\,ds$ over this surface.

As $F = xyz$, $S = x^2+y^2 = 4$, $z = 0$ to $z = 3$

$x^2+y^2-4=0$

$\mathbf{dS} = \hat{n}\,ds$ where $\hat{n} = \dfrac{\nabla S}{|\nabla S|}$

Now $\nabla S = \dfrac{\partial S}{\partial x}i + \dfrac{\partial S}{\partial y}j + \dfrac{\partial S}{\partial z}k$

$= 2xi + 2yj + 0k \Rightarrow 2xi + 2yj$

& $|\nabla S| = \sqrt{(2x)^2+(2y)^2+(0)^2} \Rightarrow \sqrt{(4x^2+4y^2)} \Rightarrow \sqrt{4(x^2+y^2)}$

$= 2\sqrt{x^2+y^2} \Rightarrow 2\sqrt{4} \Rightarrow 2(2) \Rightarrow 4$

$\therefore \hat{n} = \dfrac{\nabla S}{|\nabla S|} = \dfrac{2xi+2yj}{4} = \dfrac{2(xi+yj)}{4}$

$\hat{n} = \dfrac{xi+yj}{2}$

& $dS = \hat{n}\,ds = \dfrac{xi+yj}{2}\,ds$

$\int_S F\,ds = \int_S F\,\hat{n}\,ds \Rightarrow \int_S xyz \cdot \left(\dfrac{xi+yj}{2}\right) ds$

$= \dfrac{1}{2}\int_S (x^2yzi + xy^2zj)\,ds$

NOTES BY KINZA BILAL

Scanned with CamScanner

As $x = r\cos\theta$, $y = r\sin\theta$, $z = z$

$x = 2\cos\theta$, $y = 2\sin\theta$

Now

$$\int_S F\,ds = \frac{1}{2}\int_S (x^2yz\,i + xy^2z\,j)\,ds$$

$$= \frac{1}{2}\int_S \left[(2\cos\theta)^2(2\sin\theta)z\,i + (2\cos\theta)(2\sin\theta)^2 z\,j\right] ds$$

$$= \frac{1}{2}\int_S \left[8\sin\theta\cos^2\theta\, z\,i + 8\sin^2\theta\cos\theta\, z\, j\right] r\,d\theta\,dz$$

$$= \frac{1}{2}\cdot 8\int_0^{\pi/2}\int_0^3 \left[\cos^2\theta\sin\theta\, i + \sin^2\theta\cos\theta\, j\right] z\,d\theta\,dz$$

here $0 \le \theta \le \frac{\pi}{2}$ (bcz first octant).

$$= 4\int_0^{\pi/2} (\cos^2\theta\sin\theta\, i + \sin^2\theta\cos\theta\, j)\left[\frac{2z^2}{2}\right]_0^3 d\theta$$

$$= 4\int_0^{\pi/2} (\cos^2\theta\sin\theta\, i + \sin^2\theta\cos\theta\, j)\, 9\,d\theta$$

$$= 36\left[\left(-\frac{\cos^3\theta}{3}\, i + \frac{\sin^3\theta}{3}\, j\right)\right]_0^{\pi/2}$$

$$= \frac{36}{3}\left[\left(-\cos^3\frac{\pi}{2}\, i + \sin^3\frac{\pi}{2}\, j\right) - \left(-\frac{\cos^3 0}{\cdot}\, i + \sin^3(0)\, j\right)\right]$$

$$= \frac{36}{3}\left[j - (-i)\right] \Rightarrow 12(i + j)$$ Ans.

NOTES BY KINZA BILAL

Scanned with CamScanner

Vector Field:-

A vector field $F = yi + 2j + k$ exists over a surface S defined by $x^2 + y^2 + z^2 = 9$ bounded by $x = 0, y = 0$ $z = 0$ in first octant.

Evaluate $\int_S F \cdot dS$ over surface indicated.

$$dS = \hat{n}\, ds \quad , \quad \hat{n} = \frac{\nabla S}{|\nabla S|}$$

$$S = x^2 + y^2 + z^2 = 9$$

$$x^2 + y^2 + z^2 - 9 = 0$$

$$\nabla S = \frac{\partial S}{\partial x} i + \frac{\partial S}{\partial y} j + \frac{\partial S}{\partial z} k \Rightarrow 2xi + 2yj + 2zk$$

$$|\nabla S| = \sqrt{(2x)^2 + (2y)^2 + (2z)^2} \Rightarrow \sqrt{4x^2 + 4y^2 + 4z^2}$$

$$= \sqrt{4(x^2 + y^2 + z^2)} \Rightarrow \sqrt{4(9)} \Rightarrow \sqrt{36} \Rightarrow 6$$

$$\hat{n} = \frac{\nabla S}{|\nabla S|} = \frac{2xi + 2yj + 2zk}{6} \Rightarrow \frac{2(xi + yj + zk)}{6}$$

$$= \frac{xi + yj + zk}{3}$$

$$\hat{n}\, ds = \frac{1}{3}(xi + yj + zk)\, ds$$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$\int_S F \cdot ds = \int_S F \cdot \hat{n}\, ds = \int_S (yi + 2j + k) \cdot \frac{1}{3}(xi + yj + zk)\, ds$$

$$= \frac{1}{3}\int_S (xy + 2y + z)\, ds$$

converting into spherical coordinates

$x = r\sin\phi\cos\theta$, $y = r\sin\phi\sin\theta$, $z = r\cos\phi$

$x = 3\sin\phi\cos\theta$, $y = 3\sin\phi\sin\theta$, $z = 3\cos\phi$

$ds = r^2 \sin\phi\, d\phi\, d\theta \Rightarrow 9\sin\phi\, d\phi d\theta$

Now

$$= \frac{1}{3}\int_S (xy + 2y + z)\, ds$$

$$= \frac{1}{3}\int_S \left[(3\sin\phi\cos\theta)(3\sin\phi\sin\theta) + 2(3\sin\phi\sin\theta) + 3\cos\phi\right] 9\sin\phi\, d\phi d\theta$$

$$= \frac{1}{3}\int_S \left[9\sin^2\phi\sin\theta\cos\theta + 6\sin\phi\sin\theta + 3\cos\phi\right] 9\sin\phi\, d\phi d\theta$$

$$= 9\cdot\frac{3}{3}\int_0^{\pi/2}\int_0^{\pi/2}\left[3\sin^3\phi\sin\theta\cos\theta + 2\sin^2\phi\sin\theta + \sin\phi\cos\phi\right] d\phi\, d\theta$$

As we know, by Wallis formula.

$$\int_0^{\pi/2} \sin^3\phi\, d\phi = \frac{n-1}{n} = \frac{2}{3}$$

$$\& \int_0^{\pi/2} \sin^2\phi\, d\phi = \frac{n-1}{n}\cdot\frac{\pi}{2} = \frac{1}{2}\frac{\pi}{2}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

So we get

$$= 9\int_0^{\pi/2}\left(3\cdot\frac{2}{3}\sin\theta\cos\theta + 2\cdot\frac{1}{2}\cdot\frac{\pi}{2}\sin\theta + \int_0^{\pi/2}\frac{2\sin\varphi\cos\varphi}{2}d\varphi\right)d\theta$$

$$= 9\int_0^{\pi/2}\left(\sin 2\theta + \frac{\pi}{2}\sin\theta + \frac{1}{2}\int_0^{\pi/2}\sin 2\varphi\ d\varphi\right)d\theta$$

$$= 9\int_0^{\pi/2}\left(\sin 2\theta + \frac{\pi}{2}\sin\theta + \frac{1}{2}\left[\frac{\cos 2\varphi}{2}\right]_0^{\pi/2}\right)d\theta$$

$$= 9\int_0^{\pi/2}\left(\sin 2\theta + \frac{\pi}{2}\sin\theta + \frac{1}{4}\left[-\cos\pi-(-\cos 0^\circ)\right]\right)d\theta$$

$$= 9\int_0^{\pi/2}\left(\sin 2\theta + \frac{\pi}{2}\sin\theta + \frac{1}{4}\left(-(-1)-(-1)\right)\right)d\theta$$

$$= 9\int_0^{\pi/2}\left(\sin 2\theta + \frac{\pi}{2}\sin\theta + \frac{1}{2}\right)d\theta$$

$$= 9\left[\frac{-\cos 2\theta}{2} + \frac{\pi}{2}(-\cos\theta) + \frac{1}{2}\theta\right]_0^{\pi/2}$$

$$= 9\left[\frac{-\cos\pi}{2} - \frac{\pi}{2}\cos\frac{\pi}{2} + \frac{1}{2}\left(\frac{\pi}{2}\right)\right] - \left[\frac{-\cos 0^\circ}{2} - \frac{\pi}{2}\cos 0^\circ + \frac{1}{2}(0)\right]$$

$$= 9\left[\frac{1}{2} - 0 + \frac{\pi}{4}\right] - \left[-\frac{1}{2} - \frac{\pi}{2} + 0\right]$$

$$= 9\left[\frac{1}{2} + \frac{\pi}{4} + \frac{1}{2} + \frac{\pi}{2}\right] \Rightarrow 9\left[\frac{2+\pi+2+2\pi}{4}\right] \Rightarrow 9\left(\frac{4+3\pi}{4}\right)$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Example:- Evaluate $\int_S F.ds$ where

$F = 2yj + zk$ & S is surface $x^2+y^2=4$ in first two octants bounded by the planes $z=0$, $z=5$ & $y=0$

$$S = x^2+y^2=4$$
$$x^2+y^2-4=0$$

$$\hat{n} = \frac{\nabla S}{|\nabla S|}$$

$$\nabla S = \frac{\partial S}{\partial x} i + \frac{\partial S}{\partial y} j + \frac{\partial S}{\partial z} k$$

$$= 2xi + 2yj$$

$$|\nabla S| = \sqrt{(2x)^2+(2y)^2} = \sqrt{4x^2+4y^2} = \sqrt{4(x^2+y^2)}$$

$$= \sqrt{4(4)} = \sqrt{16} = 4$$

$$\hat{n} = \frac{\nabla S}{|\nabla S|} = \frac{2xi+2yj}{4} = \frac{2(xi+yj)}{4} \Rightarrow \frac{xi+yj}{2}$$

$$\int_S F.\hat{n}\,ds = \int_S (2yj+zk).\left(\frac{xi+yj}{2}\right)$$

$$= \frac{1}{2}\int_S (2y^2)\,ds$$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$= \frac{2}{2}\int_S y^2\,dS = \int_S y^2\,dS$$

Now $x = r\cos\theta$ & $y = r\sin\theta$, $z = z$, $dS = r\,d\theta\,dz$

$x = 2\cos\theta$ & $y = 2\sin\theta$ $= 2\,d\theta\,dz$

$$\int_S F\cdot dS = \int_S (2\sin\theta)^2\, 2\,d\theta\,dz$$

$$= \int_S 8\sin^2\theta\,d\theta\,dz \Rightarrow 8\int_S \frac{1-\cos 2\theta}{2}\,d\theta\,dz$$

$$= 4\int_0^5\int_0^{\pi}(1-\cos 2\theta)\,d\theta\,dz$$

$$= 4\int_0^5\left[\theta - \frac{\sin 2\theta}{2}\right]_0^{\pi} dz$$

$$= 4\int_0^5\left[\left(\pi - \frac{\sin 2\pi}{2}\right) - \left(0 - \frac{\sin 2(0)}{2}\right)\right] dz$$

$$= 4\int_0^5 \pi\,dz \Rightarrow 4\pi\,[z]_0^5$$

$$= 4\pi[5-0) \Rightarrow 20\pi \quad \underline{\text{Ans.}}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Conservative Vector Field:-

For a closed path in a conservative field

$$\oint_C F \cdot dr = 0$$

if field is conservative

$$\int_{C_1(AB)} F \cdot dr = \int_{C_2(AB)} F \cdot dr$$



NOTES BY KINZA BILAL

Scanned with CamScanner

Lecture 38

Conservative Vector Fields

For a closed path in conservative field $\oint F \cdot dr = 0$

Example:- If $F = 2xyz\,i + x^2z\,j + x^2y\,k$, evaluate the line integral $\int F \cdot dr$ b/w $A(0,0,0)$ & $B(2,4,6)$

(a) along the curve C whose parametric equations are $x = u,\ y = u^2,\ z = 3u$.

(b) along the three straight lines

C_1 $(0,0,0)$ to $(2,0,0)$

C_2 $(2,0,0)$ to $(2,4,0)$

C_3 $(2,4,0)$ to $(2,4,6)$

(a) $F = 2xyz\,i + x^2z\,j + x^2y\,k$

$x = u,\ y = u^2,\ z = 3u$

$dx = du,\ dy = 2u\,du,\ dz = 3\,du$

$$F \cdot dr = (2xyz\,i + x^2z\,j + x^2y\,k) \cdot (dx\,i + dy\,j + dz\,k)$$

$$= 2xyz\,dx + x^2z\,dy + x^2y\,dz$$

put $x = u,\ y = u^2$ & $z = 3u$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$F.dr = 2(u)(u^2)(3u)(du) + (u)^2(3u^2)\,2u\,du + (u)^2(u^2)\,3du$$

$$= (6u^4 + 6u^4 + 3u^4)\,du$$

$$= 15u^4\,du$$

$$\int_0^2 F.dr = \int_0^2 15u^4\,du =$$

$$= 15\int_0^2 u^4\,du$$

$$= 15\left[\frac{u^5}{5}\right]_0^2$$

$$= \frac{15}{5}\left[2^5 - 0^5\right]$$

$$= 3(32)$$

$$= 96 \text{ Ans.}$$

$x = u$, $y = u^2$, $z = 3u$

put $A(0,0,0)$ for lower limit

$0 = u$, $0 = u^2$, $0 = 3u$

$0 = u$, $0 = u$

put $B(2,4,6)$ for upper limit

$2 = u$, $4 = u^2$, $6 = 3u$

$2 = u$, $2 = u$

(b) C_1: along x-axis $(0,0,0)$ to $(2,0,0)$

put $y=0$, $z=0$, $dy=0$, $dz=0$

$$\int F.dr = \int (2xyz\,dx + x^2z\,dy + x^2y\,dz)$$

$$= 0 + 0 + 0 \Rightarrow 0$$

NOTES BY KINZA BILAL

Scanned with CamScanner

C_2: along y-axis $(2,0,0)$ to $(2,4,0)$

put $x=2$, $z=0$, $dx=0$, $dz=0$

$$\int_{C_2} F \cdot dr = \int_{C_2} (2xyz\,dx + x^2 z\,dy + x^2 y\,dz)$$

$$= 0+0+0 \Rightarrow 0$$

C_3: along z-axis $(2,4,0)$ to $(2,4,6)$

$x=2$, $y=4$, $dx=0$, $dy=0$

$$\int_{C_3} F \cdot dr = \int_{C_3} (2xyz\,dx + x^2 z\,dy + x^2 y\,dz)$$

$$= \int_{C_3} [2(2)(4)z(0) + (2)^2(z)(0) + (2)^2(4)dz)$$

$$= \int_0^6 16\,dz \Rightarrow 16[z]_0^6 \Rightarrow 16(6-0) \Rightarrow 96$$

Now

$$\int_{C_1+C_2+C_3} F \cdot dr = 0+0+96 \Rightarrow 96$$

here the value of line integral is independent of two paths. So F is a conservative field.

NOTES BY KINZA BILAL

Scanned with CamScanner

$$\text{Curl } F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2xyz & x^2z & x^2y \end{vmatrix}$$

$$= i(x^2 - x^2) + -j(2xy - 2xy) + k(2xz - 2xz)$$

$$= 0 - 0 + 0 \Rightarrow 0$$

So There are three test to check whether a vector field is conservative or not.

(a) $\oint F.dr = 0$

(b) curl $F = 0$

(c) $F = \text{grad } V$

Divergence Theorem:- (Gauss' Theorem).

For a closed surface S, with region V & vector field F

$$\int_V \text{div } F dV = \int_S F.dS$$

generally, volume integral (Tripple integral) can be expressed as a surface integral (double integral).

NOTES BY KINZA BILAL

Scanned with CamScanner

Example:- Verify the divergence Theorem for the vector field $F = x^2 i + zj + yk$ taken over the region bounded by planes $z=0$, $z=2$, $x=0$, $x=1$, $y=0$, $y=3$.

$dV = dx\,dy\,dz$.

We have to show that

$$\int_V \text{div } F\, dV = \int_S F \cdot dS$$

So $\text{div } F = \nabla \cdot F = \left(\frac{\partial}{\partial x} i + \frac{\partial}{\partial y} j + \frac{\partial}{\partial z} k\right) \cdot (x^2 i + zj + yk)$

$$= 2x + 0 + 0 \Rightarrow 2x$$

$$\int_V \text{div } F\, dV = \int_V 2x\, dV = \int_0^1 \int_0^3 \int_0^2 2x\, dz\, dy\, dx$$

$$= \int_0^1 \int_0^3 [2xz]_0^2 dy\, dx \Rightarrow \int_0^1 \int_0^3 2x(2-0) \Rightarrow \int_0^1 \int_0^3 4x\, dy\, dx$$

$$= \int_0^1 4x [y]_0^3 dx \Rightarrow \int_0^1 4x(3-0)dx = \int_0^1 12x\, dx$$

$$= 12 \int_0^1 x\, dx \Rightarrow 12\left[\frac{x^2}{2}\right]_0^1 \Rightarrow 6[x^2]_0^1 \Rightarrow 6[1^2 - 0^2]$$

$$= 6 \quad \text{Ans.}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$\int_S F \cdot ds = \int_S F \cdot \hat{n}\, ds$$

Surface consists of 6 separate plane faces $S_1, S_2, S_3, S_4, S_5, S_6$. & $F = x^2 i + z j + y k$

S_1 (for z) $\quad z = 0, \quad \hat{n} = -k$

$$F = x^2 i + y k$$

$$\int_{S_1} F \cdot \hat{n}\, ds_1 = \int_0^1 \int_0^3 (x^2 i + y k)(-k)\, dy\, dx$$

$$= \int_0^1 \int_0^3 (-y)\, dy\, dx \Rightarrow \int_0^1 -\left[\frac{y^2}{2}\right]_0^3 dx$$

$$= \int_0^1 -\left(\frac{3^2}{2} - \frac{0^2}{2}\right) dx = \int_0^1 -\frac{9}{2}\, dx$$

$$= -\frac{9}{2}[(x)]_0^1 \Rightarrow -\frac{9}{2}(1-0) \Rightarrow \boxed{-\frac{9}{2}}$$

S_2 (for z) $\quad z = 2, \quad \hat{n} = k$

$$F = x^2 i + 2j + y k$$

$$\int_{S_2} F \cdot \hat{n}\, ds_2 = \int_0^1 \int_0^3 (x^2 i + 2j + y k)(k)\, dy\, dx$$

$$= \int_0^1 \int_0^3 y\, dy\, dx \Rightarrow \int_0^1 \left[\frac{y^2}{2}\right]_0^3 dx$$

$$= \int_0^1 \frac{9}{2}\, dx \Rightarrow \frac{9}{2}[x]_0^1 \Rightarrow \frac{9}{2}(1-0) \Rightarrow \boxed{\frac{9}{2}}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

S_3 (for y) $\quad y = 0, \quad \hat{n} = -j$

$F = x^2 i + z j$

$$\int_{S_3} F \cdot \hat{n}\, ds_3 = \int_{S_3} (x^2 i + z j) \cdot (-j)\, ds_3$$

$$= \int_0^1 \int_0^2 -z \, dz\, dx \Rightarrow \int_0^1 -\left[\frac{z^2}{2}\right]_0^2 dx$$

$$= \int_0^1 -\left(\frac{2^2}{2} - \frac{0^2}{2}\right) dx = \int_0^1 -2 \, dx \Rightarrow -2[x]_0^1$$

$$= -2(1-0) \Rightarrow -2(1) \Rightarrow \boxed{-2}$$

S_4 (for y) $\quad y = 3, \quad \hat{n} = j$

$F = x^2 i + z k + 3k$

$$\int_{S_4} F \cdot \hat{n}\, ds_4 = \int_{S_4} (x^2 i + z k + 3k) \cdot (j)\, ds_4$$

$$= \int_0^1 \int_0^2 z \, dz\, dx \Rightarrow \int_0^1 \left[\frac{z^2}{2}\right]_0^2 dx$$

$$= \int_0^1 \left(\frac{4}{2} - 0\right) dx = \int_0^1 2 \, dx$$

$$= 2[x]_0^1 \Rightarrow 2(1-0) \Rightarrow \boxed{2}$$

NOTES BY KINZA BILAL

Scanned with CamScanner

S_5 (for x) $x = 0$, $\hat{n} = -\hat{i}$

$F = x^2\hat{i} + z\hat{j} + y\hat{k} = z\hat{j} + y\hat{k}$

$$\int_{S_5} F \cdot \hat{n}\, ds_5 = \int_{S_5} (z\hat{j} + y\hat{k})(-\hat{i})\, ds_5$$

$$= \int_0^2 \int_0^3 0\, dy\, dz \Rightarrow \boxed{0}$$

S_6 (for x) $x = 1$, $\hat{n} = \hat{i}$

$F = \hat{i} + z\hat{j} + y\hat{k}$

$$\int_{S_6} F \cdot \hat{n}\, ds_6 = \int_{S_6} (\hat{i} + z\hat{j} + y\hat{k}) \cdot (\hat{i})\, ds_6$$

$$= \int_0^2 \int_0^3 1\, dy\, dz \Rightarrow \int_0^2 [y]_0^3\, dz$$

$$= \int_0^2 (3-0)\, dz \Rightarrow 3\int_0^2 1\, dz$$

$$= 3[z]_0^2 \Rightarrow 3(2-0) \Rightarrow 3(2) = \boxed{6}$$

Now for whole surface

$$\int_S F \cdot ds = \frac{-9}{2} + \frac{9}{2} - 2 + 2 + 0 + 6 = 6$$

So proved That

$$\int_V \text{div } F\, dV = \int_S F \cdot dS$$

NOTES BY KINZA BILAL

Scanned with CamScanner

Stokes Theorem:- If F is a vector field existing over an open surface S & around its boundary closed curve C, then

$$\int_S \text{curl } F \cdot dS = \oint_C F \cdot dr$$

We can express a surface integral in terms of a line integral.

Example:- A hemisphere S is defined by $x^2+y^2+z^2=+4$ $(z \geq 0)$. A vector field $F = 2yi - xj + xzk$ exist over the surface & around its boundary C. verify Stoke's Theorem.

$S = x^2+y^2+z^2 = 4$, $F = 2yi - xj + xzk$

$x^2+y^2+z^2-4=0$

if we ignore z ↑ then put $z=0$ $\quad x^2+y^2=4$ (circle)

$$\oint_C F \cdot dr = \int_C (2yi - xj + xzk) \cdot (dx\,i + dy\,j + dz\,k)$$

$$= \int_C [2y\,dx - x\,dy + xz\,dz]$$

converting to polar coordinates

NOTES BY KINZA BILAL

Scanned with CamScanner

$x = r\cos\theta$, $y = r\sin\theta$, $z = 0$

$x = 2\cos\theta$, $y = 2\sin\theta$, $dz = 0$

$dx = -2\sin\theta\, d\theta$, $dy = 2\cos\theta\, d\theta$

$$\oint_C F \cdot dr = \int_0^{2\pi} \left[2(2\sin\theta)(-2\sin\theta\, d\theta) - (2\cos\theta)(2\cos\theta\, d\theta) \right]$$

$$= \int_0^{2\pi} \left(-8\sin^2\theta - 4\cos^2\theta \right) d\theta$$

$$= -4 \int_0^{\pi/2} \left(2\sin^2\theta + \cos^2\theta \right) d\theta$$

$$= -4 \int_0^{\pi/2} \left(2\sin^2\theta + 1 - \sin^2\theta \right) d\theta$$

$$= -4 \int_0^{\pi/2} \left(1 + \sin^2\theta \right) d\theta$$

$$= -4 \int_0^{\pi/2} \left(1 + \frac{1 - \cos 2\theta}{2} \right) d\theta$$

$$= -4 \int_0^{\pi/2} \left(\frac{2 + 1 - \cos 2\theta}{2} \right) d\theta$$

$$= -2 \int_0^{\pi/2} \left(3 - \cos 2\theta \right) d\theta$$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$= -2\left[3\theta - \frac{\sin 2\theta}{2}\right]_0^{2\pi} \Rightarrow$$

$$= -2\left[\left\{3(2\pi) - \frac{\sin 2(2\pi)}{2}\right\} - \left\{3(0) - \frac{\sin 2(0)}{2}\right\}\right]$$

$$= -2\left[6\pi - 0\right] \Rightarrow -2(6\pi) \Rightarrow -12\pi$$

Now $\int \text{curl } F \cdot dS = \int \text{curl } F \cdot \hat{n}\, dS.$

$$F = 2yi - xj + xzk$$

$$\text{curl } F = \begin{vmatrix} i & j & k \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ 2y & -x & +xz \end{vmatrix}$$

$$= i(0-0) + (-j)(z-0) + k(-1-2)$$

$$= -zj - 3k$$

$$\hat{n} = \frac{\nabla S}{|\nabla S|}$$

$$\nabla S = \frac{\partial}{\partial x} Si + \frac{\partial}{\partial y} Sj + \frac{\partial}{\partial z} Sk$$

$$= 2xi + 2yj + 2zk$$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$|\nabla S| = \sqrt{(2x)^2+(2y)^2+(2z)^2}$$

$$= \sqrt{4x^2+4y^2+4z^2}$$

$$= \sqrt{4(x^2+y^2+z^2)}$$

$$= 2\sqrt{4(4)} = \sqrt{16} = 4$$

So $\hat{n} = \dfrac{\nabla S}{|\nabla S|} = \dfrac{2xi + 2yj + 2zk}{4}$

$$= \frac{2(xi+yj+zk)}{4}$$

$$= \frac{xi+yj+zk}{2}$$

Now

$$\int_S \text{curl } F \cdot \hat{n}\, ds = \int_S (-3j-3k)\cdot\left(\frac{xi+yj+zk}{2}\right) ds$$

$$= \frac{1}{2}\int_S (-yz-3z)\, ds$$

$x = r\sin\phi\cos\theta$, $y = r\sin\phi\sin\theta$, $z = r\cos\phi$

$x = 2\sin\phi\cos\theta$, $y = 2\sin\phi\sin\theta$, $z = 2\cos\phi$

& $dS = r^2\sin\phi\, d\phi\, d\theta$

$= 4\sin\phi\, d\phi\, d\theta$

NOTES BY KINZA BILAL

Scanned with CamScanner

$$\int_S \text{curl } F \cdot \hat{n}\, dS = \frac{1}{2}\int_S \left[-2\sin\varphi\sin\theta\, 2\cos\varphi - 3.\ 2\cos\varphi\right] ds$$

$$= \frac{1}{2}\int_0^{2\pi}\int_0^{\pi/2}\left[-4\sin\theta\sin\varphi\cos\varphi - 6\cos\varphi\right] 4\sin\varphi\, d\varphi\, d\theta$$

$$= 4\left(\frac{-2}{2}\right)\int_0^{2\pi}\int_0^{\pi/2}\left[2\sin^2\varphi\sin\theta\cos\varphi + 3\sin\varphi\cos\varphi\right] d\varphi\, d\theta$$

$$= -4\int_0^{2\pi}\left[\frac{2\sin\theta\,\sin^3\varphi}{3} + 3\,\frac{\sin^2\varphi}{2}\right]_0^{\frac{\pi}{2}} d\theta$$

$$= -4\int_0^{2\pi}\left[\frac{2\sin\theta\,\sin^3\frac{\pi}{2}}{3} + 3\,\frac{\sin^2\frac{\pi}{2}}{2}\right] - 0 \quad d\theta$$

$$= -4\int_0^{2\pi}\left(\frac{2}{3}\sin\theta + \frac{3}{2}\right) d\theta$$

$$= -4\left[-\frac{2}{3}\cos\theta + \frac{3}{2}\theta\right]_0^{2\pi}$$

$$= -4\left[-2\,\frac{\cos(2\pi)}{3} + \frac{3}{2}(2\pi)\right] - \left[\frac{-2\cos 0}{3} + \frac{3}{2}(0)\right]$$

$$= -4\left[-\frac{2}{3} + 3\pi + \frac{2}{3}\right] \Rightarrow -4(3\pi)$$

$$= -12\pi \qquad \text{So } \int_S \text{curl } F \cdot dS = \oint_C F \cdot dr$$

NOTES BY KINZA BILAL

Scanned with CamScanner